

1st National Conference



Nepal Shelter Housing and Settlements Forum
Kathmandu, 25-26 April 2023

2023

**From Recovery to Resilience:
Implementing Nepal's Declaration on
Green, Resilient and Inclusive Development Pathway**



**Proceeding
Report**



Government of Nepal
Ministry of Home Affairs
National Disaster Risk Reduction and Management Authority
Singhadurbar, Kathmandu

1st National Conference



Nepal Shelter Housing and Settlements Forum
Kathmandu, 25-26 April 2023

Theme:

From Recovery to Resilience: Implementing Nepal's Declaration on Green, Resilient, and Inclusive Development Pathway

Nepal Shelter Housing and Settlements Forum 2023
Proceeding Report

#NSHSF2023

1st National Conference



Nepal Shelter Housing and Settlements Forum
Kathmandu, 25-26 April 2023

From Recovery to Resilience: Implementing Nepal's Declaration on Green, Resilient, and Inclusive Development Pathway

Organized by:



Government of Nepal
Ministry of Home Affairs
National Disaster Risk Reduction and Management Authority
Singhadurbar, Kathmandu

Supported by:



Save the Children



#NSHSF2023

Nepal Shelter Housing and Settlements Forum 2023
Proceeding Report

About the event

Nepal Shelter Housing and Settlements Forum (NSHSF) 2023, steered with the theme **“From Recovery to Resilience: Implementing Nepal’s Declaration on Green, Resilient, and Inclusive Development Pathway”** concluded in Kathmandu Nepal from 25-26 April 2023. The first edition of the national conference on NSHSF 2023 under the leadership of NDRRMA was organized on the Memorial Day of Gorkha Earthquake 2015 and inaugurated by Hon. Minister of Urban Development, Ms. Sita Gurung and coordinated jointly by the organizing committee members working in disaster management and humanitarian sector.

The forum was supported largely by actors and stakeholders from three tiers of government, from federal to provincial and local along with a bunch of development partners and I/NGOs that played a major role in providing humanitarian shelter and settlements support to people affected by disasters in Nepal including of the 2015 Gorkha earthquake. The conference also saw active participation from CSOs, academics, researchers, private sectors and interested ones having common perspectives in DRRM, disaster resilience and recovery.



This marvelous forum highlighted the need for enhanced coordination and collaboration among the three tiers of government (federal, provincial, local) and with humanitarian system to enhance disaster preparedness and response, reduce multi-hazard disaster risks, and promote localized, green, resilient, and inclusive disaster recovery efforts. The participating local government identified their role as the front-liners, the provincial government acknowledged their status being in early establishment phase, the federal government highlighted the need of translation of policies to provincial and local level whereas the humanitarian agencies shared their best practices and impact working with communities and government.

1st Nepal Shelter Housing and Settlements Forum 2023 Proceeding Report

July 2023



Acknowledgement

Different level of valuable stakeholders and personnel whose coordination and insights have supported the Nepal Shelter Housing and Settlements Forum 2023 (#NSHSF 2023) to achieve its success. NSHSF Secretariat team is thankful to representatives of all identified levels of government stakeholders, development actors, private sectors, and academics for sharing their useful inputs and guidance throughout several preparatory meetings of NSHSF Organizing Committee as well as the active participation during the main event days.

NSHSF Organizing Committee Team along with the NSHSF Secretariat Organization (NHSRP) is indebted towards the leadership of NDRRMA and most primarily the Chief Executive, Mr. Anil Pokhrel for sharing valuable program experiences, inputs, mentoring and valuable way forward for event design and in overall event organization. We would like to express heartiest thanks towards the Project Director of CLPIU (Building & Housing), Mr. Koshnath Adhikari for assuring the presence of Hon- Minister of Urban Development, Ms. Sita Gurung during the opening session in which deliberate coordination efforts were laid by the NHSRP, Secretariat of NSHSF 2023. We share the immense gratitude towards all

concerned stakeholders of Ministry of Home Affairs (MoHA)/Government of Nepal particularly to Joint Secretary, Mr. Pradip Koirala for the support in guidance towards the opening session and during the panel session on provincial governments. We are also thankful to Council Member of NDRRMA, Dr. Ganga Lal Tuladhar for his valuable insights during the closing session.

We extend our sincere thanks to the Organizing Committee team members for ensuring the role of Session Moderators during the couple of parallel and panel sessions. The Moderators representing NDRRMA were Mr. Anil Pokhrel, Mr. Rameshwor Marahatta, Mr. Rudra Singh Tamang, Mr. Krishna Gaire and Mr. Sundar Prasad Sharma. The Moderators representing MoFAGA was Mr. Rishi Raj Acharya, from MoLCPA was Mr. Chandra Bahadur Shiwakoti and from CLPIU (Building & Housing) was Mr. Koshnath Adhikari. All the Panel Members deserves sincere thanks from NSHSF Organizing Committee team. We would also like to share our sincere thanks to Former Chief Executive Officer of NRA, Mr. Sushil Gyewali for enduring efforts in sharing the experiences of 2015 Nepal reconstruction and providing a way forward during a Panel Session.

We would like to sincerely thank Panellists of all the Panel Sessions and indebted towards their insights that they provided. Similarly, we extend our sincere thanks to representatives of Provincial Ministries mainly Mr. Dhurba Gaida-Secretary, MoIAL (Bagmati), Mr. Umesh Basnet-Secretary, MoIAL (Koshi) Mr. Suran Prasad Sharma Paudel-Under Secretary, MoIAL (Gandaki), Mr. Nidhi Raj Neupane-Under Secretary, MoIAL (Karnali), Mr. Dharma Raj Ojha-Senior Land Survey Officer, Ministry of Land Management, Agriculture and Cooperative (Sudurpaschim) as their valuable insights during the Panel Session. We are also thankful towards the members from MuAN (Mr. Kalanidhi Devkota, Executive Director) and NARMIN (Ms. Laxmi Devi Pande, Chairperson) and the elected representatives from local levels (Mr. Jit Narayan Thapa Magar, Budhiganga Rural Municipality; Mr. Jaya Narayan Shah, Jhapa Rural Municipality) and Mr. Bishnu Paudel (Tansen Municipality) for their valuable sharing in the Panel Session.

NSHSF2023 is not possible without the active support from Organizing Committee members who carried multiple roles and laid technical assistance throughout the event from pre-event to during-event phase. NSHSF Secretariat is thankful towards the support provided by its staff members and volunteers' squad during the main event days; the main team in the information technology management, communication, reporting and overall, the administration

and logistics management, the team at the registration desk and last but not the least the bunch of volunteers supporting in the notes taking efforts.

It is with the generous financial support from NSHSF Organizing Committee members comprising of USAID Nepal, UKAid Nepal, Catholic Relief Services (CRS), Habitat for Humanity Nepal (HfH), International Federation of Red Cross and Red Crescent Societies (IFRC), Nepal Red Cross Society (NRCS), International Organization of Migration (IOM), National Society for Earthquake Technology Nepal (NSET), People in Need Nepal (PIN), Save the Children Nepal, CARE Nepal and Plan International Nepal to prepare the Terms of Reference (ToR), design the program modality and work on the agenda along with the arrangements of related administrative and operations procedures.

Abbreviations

ADPC	Asian Disaster Preparedness Centre
ASF	Asia Shelter Forum
BPfA	Beijing Declaration and Platform for Action
CBFT	Cement Bamboo Frame Technology
CEDAW	Convention on the Elimination of All Forms of Discrimination against Women
CLPIU	Central Level Project Implementation Unit
CNNC	City Net National Chapter
CRS	Catholic Relief Services
CSEB	Compressed Stabilized Earth Brick
CSOs	Civil Society Organizations
DHM	Department of Hydrology and Meteorology
DoLI	Department of Local Infrastructure
DPC	Damp-Proof Concrete
DRR	Disaster Risk Reduction
DRRM	Disaster Risk Reduction and Management
DUDBC	Department of Urban Development and Building Construction
ERPP	Emergency Response and Preparedness Plan
EWS	Early Warning System
FGDs	Focused Group Discussions
GEDSI	Gender, Disability and Social Inclusion
GEM	Global Earthquake Model in Nepal
GoN	Government of Nepal
GRDC	Global Research and Development Center
GRID	Green Resilient Inclusive Development
HfH	Habitat for Humanity Nepal
HIWAT	High-Impact Weather Assessment Toolkit
HVRCA	Hazard, Vulnerability, Risk and Criticality Assessment
I/NGOs	International and/National Non-Governmental Organizations
ICIMOD	International Centre for Integrated Mountain Development
IFRC	International Federation of Red Cross and Red Crescent Societies
IoE	Institute of Engineering

IOM	International Organization of Migration
IRA	Individual Roof Area
MoFAGA	Ministry of Federal Affairs and General Administration
MoHA	Ministry of Home Affairs
MoLCPA	Ministry of Land Cooperatives and Poverty Alleviation
MoUD	Ministry of Urban Development
MuAN	Municipal Association of Nepal
NARMIN	National Association of Rural Municipalities in Nepal
NDRF	National Disaster Resilience Framework
NDRRMA	National Disaster Risk Reduction and Management Authority
NHSRP	National Housing and Settlement Resilience Platform
NPR/NRs	Nepali Rupees
NRA	National Reconstruction Authority
NRCS	Nepal Red Cross Society
NSET	National Society for Earthquake Technology Nepal
NSHSF	Nepal Shelter Housing and Settlements Forum
PDNA	Post Disaster Needs Assessment
PIN	People in Need Nepal
PMEP	Prime Minister Employment Program
RADO	Rural Awareness and Development Organization
RSLUP	Risk Sensitive Land Use Planning
SPCMA	Shelter and Housing Pre-Crisis Market Analysis
SPH	Sindh Peoples Housing for Flood Affects
SSA	Social Security Allowance
ToT	Training of Trainers
ToR	Terms of Reference
UKAid	UK Department for International Development
UNCRPD	United Nations Convention on the Rights of Persons with Disabilities
USAID	United States Agency for International Development
USD	United States Dollars
WHDRRP	Women Humanitarian and DRR Platform

Table of Contents

1. Executive Summary	1
2. Introduction	5
2.1 Participants	6
2.2 Program Outline	6
2.3 Objective	8
3. Welcome and Inauguration Session	9
3.1 Welcome	9
3.2 Opening Session	10
4. Remarks	11
5. Testimonials	15
6. Key Presentation	18
7. Key Parallel Sessions - Day 1	21
7.1 DRRM Governance, Multi-Hazard DRR, and Climate Change Adaptation	21
7.2 Learning from post-earthquake reconstruction: How localization worked and what were the gaps?	25
7.3 Multi-hazard DRR and Climate change Adaptation	28
7.4 To the Core of the Localization: Working with people, communities, and local government	32
8. Key Parallel Sessions - Day 2	37
8.1 Multi-Hazard Risk Assessment	37
8.2 Integrating Gender Disability and Social Inclusion (GEDSI)	41
8.3 Greening Shelter and Sustainable Housing Practices	45
8.4 Affordable Housing Solutions with Innovative Practices	49
8.5 Integrating WASH, Livelihoods, Land for Building Resilient Homes and Communities	53
8.6 Multi Hazard Early warning System	57
9. Key Panel Sessions	61
9.1 Lesson Learned & Way Forward for a Green, Multi-Hazard Resilient and Inclusive Recovery	61
9.2 Lessons from 2015 Earthquake Recovery	66
9.3 Provincial Government Pathway towards Resilience	70
9.4 Local Government Pathway towards Resilience	79
10. Closing Session	84
10.1 Appreciation and Token of Love distribution	84
10.2 Closing Remarks by NSHSF Secretariat and Organizing Committee	87
10.3 Closing Remarks from Council Member of NDRRMA, Dr. Ganga Lal Tuladhar	88
10.4 Closing Remarks from Chief Executive of NDRRMA, Mr. Anil Pokhrel	89
11. Participants Feedbacks and Reflection on NSHSF	90
12. Summary of Recommendations	92
Annexes	95



1. Executive Summary

The Nepal Shelter Housing and Settlements Forum (NSHSF) 2023, held in Kathmandu Nepal from 25-26 April 2023 marked the Memorial Day of Gorkha Earthquake 2015. The forum, guided by the theme “From Recovery to Resilience: Implementing Nepal’s Declaration on Green, Resilient, and Inclusive Development Pathway (GRID)” was successfully organized under the leadership of the National Disaster Risk Reduction and Management Authority (NDRRMA). The opening ceremony was inaugurated by the Honorable Minister of Urban Development (MoUD), Ms. Sita Gurung.

The forum was supported largely by actors and stakeholders from three

tiers of government, from federal to provincial and local along with a bunch of development partners and international/non-governmental organizations (I/NGOs) that played a major role in providing humanitarian shelter and settlements support to people affected by disasters in Nepal including of the 2015 Gorkha earthquake. The conference also saw active participation from Civil Society Organizations (CSOs), academics, researchers, private sectors and interested ones having common perspectives in Disaster Risk Reduction and Management, disaster resilience and recovery.

The concluded commitment from two-days NSHSF conference projected the

requirement of Green Resilient and Inclusive Development (GRID) incorporation at every possible sector to make the dream reality of Nepal as a resilient nation by 2030. The participants reiterated and reinforced for commitments geared towards improved coordination and collaboration between three tiers of government for the sound and effective disaster preparedness & response, disaster risk reduction and green, resilient, and inclusive disaster recovery.

A total of four panel sessions and forty-five thematic presentations covering several cross-cutting themes and innovations on shelter and housing reconstruction and DRRM initiatives were shared by speakers, representing sectoral governmental agencies, partner organizations and researchers.



Majority of speakers in the parallel sessions reflected their organizational and research experiences of working at the community level in collaboration of diverse governmental agencies (from ministries to departments, districts, and

local governments). On top of this, the speakers working in frontline thematic sectors illustrated best practices and proven achievements from thematic sectors of shelter, housing, settlements, recovery, reconstruction and DRR, disaster preparedness, climate adaptive affordable green solutions and many more compelling ideas. Not only this, the speakers attempted to draw attention of diverse participants about the challenges and gaps and bottlenecks issues and sought the recommendations also.

Maximum parallel sessions reiterated over localization of the policies, finding the localized solutions, capacity enhancement and sensitization of the public and targeted groups of community members (women, youths, farmers, people with disabilities etc.) for their inclusive participation, stressed for proactive role of local bodies and data driven decision making. Proper coordination among three tiers of government, with development partners and CSOs were emphasized to reduce and manage and face the disaster from recovery to resilience.

Joint presentation by NHSRP with representatives of Sindh Peoples Housing for Flood Affectees (SPHF), Government of Pakistan focused on comparing the commonality in the general situation of Nepal and Pakistan after the disasters that both nations faced in 2015 and 2022 respectively. The joint presentation

emphasized that regional collaboration is the key to tackle cross border initiatives on disaster recovery and climate change adaptation.



The discussions over clarity over work division and resource management among three tiers of government mainly skyrocketed in the question and answering session where participants provided feedback over timely, inclusive, and localized solutions from federal governmental agencies towards the local bodies. Capacity development of local level including elected members for proper functionalization of everyday tasks, development and implementation of policies and laws (including model laws and model guidelines) was also suggested.



Provincial ministries Panelists stressed that disaster is a cross cutting issue therefore requested for appropriate budget planning as well as deployment of skilled human resources on DRRM at provincial & local level. The Panelists reiterated the roles of federal governments in mobilizing the further coordination with provincial governments for effective DRRM.

The sharing by local government Panelists centralized on coordination and networking among the three tiers of governments while reiterated DRRM as the common shared responsibility and highlighted that local governments are the front liners in the disaster, therefore urged for inter-municipal cooperation for smooth disaster response, preparedness, and disaster recovery initiatives.

As a takeaway's messages, the renowned experts along with the participants at large expressed that as Nepal has signed declaration on GRID mechanisms to integrate in all possible programmatic interventions, the post-earthquake reconstruction and rehabilitation learnings along with the disaster recovery interventions can be linked in implementation of GRID pathways to achieve effective disaster recovery and resilience community level. The participants held an opinion that platforms such as NSHSF can be more effective every year to review and reflect over the majority of DRRM related initiatives and to take

forward the learnings in most effective ways. On top of this, it was suggested that events like NSHSF should be localized at provincial and local level too for assuring localization of the policies and program implementation.

NSHSF was envisioned through vigorous discussions from Asia Shelter Forum (ASF) International Conference which was organized in November 2021 in Kathmandu Nepal. It was agreed among all concerned governmental and non-governmental shelter, settlements and DRRM practitioners to hold an annual event - 'NSHSF' and work towards institutionalizing the engagement by building on existing coordination mechanisms.

National Housing and Settlements

Resilience Platform (NHSRP)/the former Housing Recovery and Reconstruction Platform -HRRP) successfully taken responsibility as the Secretariat of NSHSF's Organizing Committee Members including of USAID Nepal, UKAid Nepal, Catholic Relief Services (CRS), Habitat for Humanity Nepal (HfH), International Federation of Red Cross and Red Crescent Societies (IFRC), Nepal Red Cross Society (NRCS), International Organization of Migration (IOM), National Society for Earthquake Technology Nepal (NSET), People in Need Nepal (PIN), Save the Children Nepal, CARE Nepal and Plan International Nepal to prepare the Terms of Reference (ToR), design the program modality and work on the agenda along with the arrangements of related administrative and operations procedures.





2. Introduction

This report consolidates the proceedings of NSHSF 2023 and includes the major highlights of the panel discussions and parallel presentations during different thematic learning and sharing sessions, and the experience of experts and guests, in the forms of remarks and way forward. The conference highlighted the importance of having a common networking and sharing forum such as NSHSF to facilitate collaboration and cooperation between shelter, housing, and disaster risk reduction & management practitioners and government agencies. Under the leadership of NDRRMA/Government of Nepal, NSHSF culminated a vibrant two-days conference highlighting the importance of clarifying disaster risk reduction and management

responsibilities among all three levels of government whereby largely focused on uplifting the homes and communities during post-disaster reconstruction and recovery through green resilient inclusive development pathways.



2.1 Participants

There were about 517 participants (453 in-person, 64 virtual). Of the total in-person attendees, 23% female and 77% male participants attended the conference. Of total virtual attendees, 34% female and 66% male participants logged in from ZOOM and Facebook live. Official inauguration of NSHSF 2023 was done by the Hon. Minister of Urban Development in presence of three tiers of government (Federal, Provincial, and Local), representatives from I/NGOs, civil societies, academics, private sector stakeholders, activists, students, and media at large. These diverse stakeholders assembled to discuss housing policy, best practices and affordable approaches and its implementation priorities along with identifying the coordination and collaboration support from all concerned sectors.

2.2 Program Outline

Following Government of Nepal's (GoN) commitments on GRID, the first ever national conference on NSHSF was driven by a main programmatic theme of "From Recovery to Resilience: Implementing Nepal's Declaration on Green, Resilient, and Inclusive Development Pathway", highlighted the need for shelter housing and settlements to be sustainable (socially, economically, and environmentally) and resilient (absorptive, adaptive, and transformative).



The four sub-themes of the NSHSF were identified in line with the current challenges and needs of the recovery, reconstruction, and resilience sector encircling GRID pathways. They are:

- i. Localized Approach
- ii. Multi-hazard DRR, Climate Change Adaptation and Strengthening DRRM Governance
- iii. Innovative, Green and Affordable Practices
- iv. Integrated Approach

The conference featured six parallel sessions and four panel sessions, each focusing on mentioned subthemes. The panel sessions consisted of distinguished experts in the fields of DRRM, disaster recovery, and resilience. Two of the panel sessions specifically involved representatives from provincial and local governments, allowing their valuable perspectives to be shared and capturing lessons learned and visionary outlooks. The parallel sessions comprised presentations followed by question-and-answer sessions, featuring contributions from international non-governmental organizations (INGOs), the private sector, and government departments.

Besides the regular sessions in two

days, market stall at NSHSF event provided an opportunity to showcase innovative approaches and best practices in the green shelter & initiatives and humanitarian response mechanisms. In the context of green shelter initiatives, the market stall showcased various environmentally friendly and sustainable practices related to shelter construction and humanitarian management. This could include demonstrating the use of disaster resilient, renewable, and green housing practices energy sources, such as solar panels or biogas systems, lime stabilized soil pilot, use of bamboo as green materials etc. The market stall displayed materials and techniques that are affordable and promotes indigenous housing solutions too.



YouTube Link: <https://youtu.be/uxS471NEEyo>



2.3 Objective

The primary objective of the NSHSF 2023 is to bring together stakeholders from the disaster recovery and shelter, housing and settlements rehabilitation and rehabilitation in Nepal, discuss rampantly and connect the issues related to these sectors towards implementing Nepal's declaration on the GRID pathways. The national event further aims at featuring wider discussion on sustainability of longer-term shelter and settlements recovery and focuses on:

- i. Creating a common forum for government agencies, I/NGO), CSOs, and other like-minded stakeholders to exchange information, share best practices, and identify solutions to emerging issues and challenges.
- ii. Providing an opportunity to showcase innovative approaches and best practices in the shelter, housing, and settlement sectors, and to recognize the achievements of researchers and organizations working in the sectors.
- iii. Playing an important role in advocating and promoting collaboration, coordination, and partnership among stakeholders along with three tiers of government in the disaster response, recovery, and reconstruction sector in Nepal.



3. Welcome and Inauguration

3.1 Welcome

On the first day of NSHSF 2023 National Conference, Ms. **Ranjana Shrestha**/ Facilitator of the welcome opening session and/the Communication Expert of NDRRMA expressed warm greetings and graciously welcomed all the invited guests and participants including primarily of all invited esteemed guests and dignitaries representing governmental agencies of Nepal, distinguished participants from donors, participants from I/NGOs, participants who were present physically and virtually, media personnel and staff members of NSHSF Secretariat and Organizing Committee at large for their participation in first edition of NSHSF 2023.





3.2 Opening Session

The conference was inaugurated by the Honourable Minister of Urban Development, Ms. Sita Gurung.

First national edition of NSHSF 2023 aimed to initiate an effective learning and networking platform for disaster recovery, risk reduction, shelter, rehabilitation practitioners and relevant stakeholders in Nepal. The forum largely intended to explore the gaps, challenges along with unpacking the opportunities and suggest a way forward by working with all concerned government agencies, from federal to provincial and local level to meet the objectives of making Nepal disaster resilient by 2030.





4. Remarks

4.1 Welcome Remarks by Joint Secretary of NDRRMA, Mr. Rameshwor Marahatta

Joint Secretary of NDRRMA, Mr. Rameshwor Marahatta welcomed all the participants and introduced the conference theme, "Recovery to Resilience: Implementing Nepal's Declaration on the Green, Resilient, and Inclusive Development Pathway". He reflected on the Gorkha earthquake 2015 and expressed his condolences to all the souls who lost their lives. In his welcome remarks, Mr. Marahatta informed the participants about the interventions of NDRRMA and

explained that NDRRMA is implementing and providing grants for resettlement, reconstruction, and relocation as per the Monsoon-induced Disaster Affected Private Housing Reconstruction and Rehabilitation Grant Disbursement Procedures 2077 (amended 2078) in collaboration with the three tiers of government.

Mr. Marahatta further elaborated that NDRRMA has received numerous requests from local authorities for geohazard assessment studies to enable the disaster affected people in relocation and resettlement in safe areas. However, due to budgetary and technical resource constraints, these requests have been difficult to fulfill. Budget constraints and

lack of technical human resources have prevented NDRRMA from completing detailed damage assessments of the Khotang and Doti earthquake too, added by Mr. Marahatta who emphasized the need for the Nepal government to prioritize the discussions on efficient procedures and policies to enable the prompt reconstruction of private housing. NDRRMA has begun working on Multi Hazard Risk Assessment guidelines and drafted 'Concept Note on Multi Hazard Early Warning and Action Plan'. Additionally, a rapid visualization screening survey of 10,700 facilities, including schools and government buildings, is currently ongoing in 145 municipalities to prepare a priority investment plan, added by Mr. Marahatta.

4.2 Remarks by Special Guest, Secretary of MoUD, Mr. Suresh Prasad Acharya

On behalf of MoUD, Special Guest and Secretary of MoUD, Mr. Suresh Prasad Acharya commenced the speech by introducing the conference theme, 'Recovery to Resilience,' and stressed on the critical value of the thematic relevant to our country, as Nepal is prone to multiple hazards, including floods and landslides and alarms for effectiveness disaster preparedness, response, and recovery.

Secretary recalled the Nepal government's reconstruction achievements after the Gorkha earthquake. Out of the 832,585 houses that were damaged either partially or completely during the earthquake in 32 districts, 88% of the reconstruction has been completed. The Nepal government has provided grants worth 232,000 million for the reconstruction of 733,214 private houses in these districts. Furthermore, he mentioned the progress on the reconstruction of health institutes, heritage sites, government buildings, office premises, retrofitting of Tri-Chandra Campus along with earthquake protection activities conducted at the President's government residence. Secretary reassured that the Department of Urban Development and Building Construction (DUDBC) and the NDRRMA jointly will collaborate to complete the remaining activities.



Secretary also stressed that to minimize the risk of future disasters, it is crucial to reconstruct private housing, buildings, and other structures in a safer manner, following the guidelines set by the National Building Code. DUDBC as the lead agency of GoN will further facilitate these efforts in collaboration with relevant agencies, particularly provincial and local governments. Furthermore, DUDBC will provide technical expertise and capacity-building training as needed. Overall, effective coordination and cooperation between the three tiers of government is essential for the success of these initiatives.

4.3 Remarks by Chief Guest, Honorable Minister of Urban Development, Ms. Sita Gurung

The Hon. Minister of MoUD, Ms. Sita Gurung recognized the contributions of the former disaster units and other leaders of the cabinet who have made significant efforts in the field of disaster management. The Minister recalls that she herself as Member of Parliament has advocated for the establishment of various authorities and policies aimed at improving disaster preparedness, mitigation, and management, which led to the creation of the current NDRRMA by Nepal Government. However,

despite having policies and departments, she acknowledges that the authority needs additional support to strengthen DRRM and preparedness interventions.

Hon. Minister Gurung emphasized the significance of developing plans for both emergency and sustainable disaster preparedness, mitigation, and management. She stressed that the authorities should focus on establishing an 'Integrated Response Unit' to provide a response within 48 hours of any calamities happening. She also expressed her expectations that the two-day conference would be a good platform to discuss innovative approaches to further improve disaster preparedness, mitigation, and management.

In earlier phases, housing modalities for reconstruction after the post-earthquake lacked a systematic approach and failed to address the needs of families from different



geographical areas. The two-room house model did not provide sufficient living space for families of 10 members and did not leave any room for storing agricultural crops. Minister Gurung further emphasized the importance of adhering to guidelines and standards of National Building Code (NBC) to ensure future resilience to disasters for all urban structures and new construction projects. Regular monitoring and evaluation of these activities with the participation of all relevant departments and authorities is crucial. As a result, this will form a strong foundation for improvements in DRRM and disaster resilience.



4.4 Closing Remarks by Session Chair and Joint Secretary of NDRRMA, Mr. Rudra Singh Tamang

While delivering closing remarks during the inauguration session, Joint Secretary Tamang emphasized that the discussions in the various technical sessions and panel discussions, attended by relevant agencies, stakeholders, and organizational partners, will remain instrumental in fostering a way forward and fulfilling the core objectives of the conference. Mr. Tamang encouraged all the participants to actively engage and participate in the discussions of these two days of dialogue.





5. Testimonials

The testimonials shared by the local level beneficiaries and government actors aimed to showcase the efforts put forward by the disaster affected household members, affected rural municipality and ward level units and provided a platform to share how the disaster response and preparedness initiatives were undertaken.

5.1 Nepal earthquake 2015 affected beneficiary from Ramechhap Municipality (Ramechhap district), Ms. Dev Kumari Pariyar

The testimonials delivered by Ms. **Dev Kumari Nepali**, resident of Babiya Kharkha,

Ward 8, Ramechhap Municipality who is also a president of Dalit Women Federation of Ramechhap district highlighted on the sufferings and experiences during the time of 2015 Gorkha earthquake.

She emphasized that it is evident that disasters have a more significant impact on marginalized and vulnerable groups, such as women, children, and people with disabilities. Since the impact of those disasters is huge, the rescue, response and health services coverage were not efficient which caused chaos and difficulties for chronic illness peoples to access medication and pregnant and lactating women maternity delivery services. Likewise, due to open defecation it created an

environment of risk for women and threat to their menstrual hygiene.

She noted that through various government and non-government organizations their community was able to build temporary shelter along with communal toilets after a while after that earthquake. Likewise, when the Nepal Government announced to provide private housing grants all of them were very happy however, to access such grants, following technical guidelines was challenging to marginalized groups. She indicates that the private housing grant itself is not sufficient for reconstruction and many beneficiaries from their community faced extreme challenges to access the 2nd tranche. Despite the government's announcement of loans with no interest, the implementation was not carried out properly. Also, she remarked that despite the grievances system, there are still few



people who have been left out and have not received the 3rd grants for reconstruction. She urged the government for effective inclusion of marginalized groups and addressed their housing problems to bounce back to their normal life with resilient livelihoods.

YouTube Link: <https://youtu.be/IvBNydOkxM4>

5.2 Chairperson of Marsyangdi Rural Municipality (Lamjung district), Mr. Arjun Gurung

The testimonials delivered by Mr. **Arjun Gurung**, Chairperson of Marsyangdi Rural Municipality (Lamjung) shared key highlights from the monsoon-induced disaster incident that occurred during



July 2020 and the efforts undertaken by the Rural Municipality Office to address recovery, reconstruction and rehabilitation needs of the affected beneficiaries.

Through describing the series of landslides incidents in Lamjung district and initiation taken by local government for the relocation process for displaced communities, he shared that the policy and procedure are not clear and unrealistic while implementing. The major bottlenecks are lengthy processes and going through all the levels of government for approval itself is time consuming when on the other side, the affected communities need quick response to continue their daily life. Likewise, he mentioned that there are still affected people who have been left out

to access the private housing grants and access the services and policies mentioned in Monsoon Preparedness and Response Plan guidelines.

Mr. Gurung emphasized the importance of similar understanding of policies and procedures among the three-tiers of governments to ensure smooth implementation and projection of support from other hands. He shared that out of the 30 relocated households, 9 households will be supported by CRS. Despite the support from CRS and the local government, there was still a limited budget available to support the relocated households. He urged all the relevant authorities for continuation of support and coordination to facilitate the beneficiaries for proper relocation.

YouTube Link: <https://youtu.be/IvBNydOkxM4>





6. Key Presentation

6.1 Pakistan Nepal Collaboration for Resilient Recovery and Climate Adaptive Solutions

National Coordinator of NHSRP, Mr. **Minar Thapa Magar** together with representative of Sindh Peoples Housing for Flood Affectees (SPHF), Government of Pakistan, Ms. **Sana A Khowja** (Communication Lead and Technical Advisor - SPHF) shared the joint learning initiatives on regional collaboration with SPHF, and its housing cash grant program. The presenters talked about Nepal-Pakistan

regional collaboration on the ground of organizational mandate, core functions, the responsibilities and evaluated the situation of Nepal and Pakistan in the context of disaster preparedness and management. They compared the commonality in the general situation of Nepal and Pakistan after the disasters that both nations faced in 2015 and 2022 respectively and discussed the possible collaboration of the two countries in resilient recovery and adaptive solutions.

Mr. Magar talked about the support aids provided by NHSRP in construction of resilient houses and retrofitting aids along with development of policies for resilient building while Mr. Sana talked about

SPHF's aids in building resilient houses including water harvesting and solar panels fitting. NHSRP's research study on urban reconstruction and retrofitting works in urban areas of Nepal, that was gathered from the partners working in urban housing recovery has elaborated the loss and recovery efforts along with the lessons learned throughout the period. Similarly, the initiatives undertaken for flood affected people in Sindh mainly focused on the solidarity towards securing justice of vulnerable groups of people.

Mr. Minar's sharing emphasized on the major strategies for the success of NHSRP's (Former HRRP) initiatives in coordination of housing recovery and reconstruction that articulated the strategies of four pillars: Transparency, Building Resilient Houses, Assuring Land Ownership to the Beneficiaries and International Collaboration. Since its establishment, NHSRP acted as a common voice for partner organizations in facilitating advocacy at national and district level to advocate for, and to facilitate for improved coverage and quality of technical assistance.

Further, NHSRP geared its initiatives towards strict adherence of building code and key engineering requirements that makes the houses resilient, and coordination was intervened through directly working with local governments and beneficiaries. Mr. Minar reiterated that international collaboration is regarded as among the best practices to tackle vulnerable

conditions in a developing nation like Nepal. Collaborations were mapped mainly by including various activities such as capacity building of local governments and communities, multisectoral responsibility for recovery and exchange of policy and guidelines that are relevant to one another.

Ms. Khowja recalls the timeline of exchange visits undertaken by the Chief Executive Officer and Board of Directors of SPHF in Nepal where NHSRP had the opportunity to present the key highlights of post-disaster housing recovery and reconstruction interventions of Nepal. As a result, a cross-border sharing and learning opportunity was offered to NHSRP and Mr. Minar that gave new horizons to exchange resilient recovery practices for both the countries having somewhat similar kind of geographical condition. Ms. Khowja proudly shared stories of successful housing recovery of beneficiaries in flood



affected areas of Sindh. Further, there is a target of reconstruction of 350,000 resilient houses of verified beneficiaries by the end of November 2023 in Sindh. On top of this, reconstruction of 2 million houses in the affected areas will be completed within the next two years.

The key takeaways of the marvellous sharing of initiatives between two countries delve into the importance of international

and regional collaboration that induces the learning and aids in promoting adaptive solutions to calamities of climate change and enhances resilient recovery practices. The conversation intended to highlight the essence of capacity building in an international arena, to bring to spotlight the possibility of inter-country exchange of relevant guidelines and policies.

YouTube Link: <https://youtu.be/pzam95zMAR8>

Presentation: <https://bit.ly/PakNepalPPt>

The image shows a presentation slide with two main sections. On the left, under the heading '2015 Earthquake in Nepal', there is a photograph of a destroyed village. Below it, a list of statistics is partially visible: Nepal ranks 4th in terms of vulnerability to climate change, 800,000 HHs impacted, 8 million people affected, and 7.6 billion USD in damages. On the right, under the heading '2022 Flood in Pakistan', there is a photograph of a flooded area with a narrow path through the water. Below it, a list of statistics is partially visible: Pakistan ranks 8th in terms of vulnerability to climate change, and over 2.1 million HHs impacted. A speaker in a light blue suit is standing in front of the slide, holding a microphone. The date '4/25/2023' is visible in the bottom left corner of the slide.

2015 Earthquake in Nepal

- Nepal ranks 4th in terms of vulnerability to climate change
- 800,000 HHs impacted
- 8 million people affected
- 7.6 billion USD in damages

2022 Flood in Pakistan

- Pakistan ranks 8th in terms of vulnerability to climate change
- Over 2.1 million HHs impacted

Photo: © UNICEF/Zaidi

4/25/2023



7. Key Parallel Session - Day 1

7.1 DRRM Governance, Multi-Hazard DRR, and Climate Change Adaptation

The parallel session aimed to showcase the efforts laid towards sound DRRM governance and recovery from multi-hazard disaster initiatives by illustrating several exemplary best practices for effective response and recovery and implementation of policies and regulations. The speakers and their presentation focused on:

- Minimum Reconstruction Guidelines for Flood Resilient Houses Safe-

Environmentally Friendly-Affordable, Mr. Muhammad Naveed, Engineering Team lead, Government of Pakistan

- Shock Responsive Social Protection in DRRM Governance, Mr. Hemanta Dagal, Save the Children Nepal
- Post-Earthquake Reconstruction and Recovery of Private Residential Buildings following Doti Earthquake, Mr. Rabin Chaulagain, NHSRP
- Enhancing Disaster Management Process in Nepal through Good Governance: Best Practices for Effective Response and Recovery”, Mr. Khagendra Awasthi, Global Research and Development Center (GRDC)

Session Chair/Moderator: Mr. **Krishna Gaire**, Under Secretary, NDRRMA

Facilitator: Mr. **Rabin Dahal**, Save the Children Nepal

Mr. **Hemanta Dangal** from Save the Children Nepal talks about the use of social protection systems to mitigate the impact of large-scale shocks that affects whole communities, regions or even with national reach and support households affected by such shocks. Mr. Dangal also highlighted the social protection system leveraging to meet their needs before, during and after shocks for quicker response and effective results in less operational cost and gave examples on the business processes of major programs which includes Social Security Allowance (SSA), Prime Minister Employment Program (PMEP). Allowances from these schemes are provided through a banking system where more than 3.7 million people are included. The presentation concluded by highlighting



that SSA is partially ready for the support of Shock Responsive Social Protection (SRSP) particularly in piggybacking and vertical expansion, however some areas of the process still need to strengthen their payment mechanisms. He highlighted that the much-awaited SRSP guideline shall address the SRSP issues like design tweaks, vertical as well as horizontal expansion.



Er. **Muhammad Naveed** represented Sindh People's Housing for Flood Affectees (SPHF), the Government of Sindh, Pakistan and his presentation mainly focused on the minimum reconstruction guideline for flood resilient houses and safe- environmentally friendly- affordable houses in the context of Pakistan. He elaborated the guiding principles of housing reconstruction, some of which included technical review and approval. inspection regime, local building materials, minimum size of the house etc. He also highlighted the proposed guideline for resilient and sustainable

housing construction that includes site selection, sub-grade parameters (soil and compaction), specification of local bearing walls foundation, floor finishing and apron/ slope protection and damp-proof course (DPC). Mr. Naveed also discussed the construction procedure of Nepal and Pakistan and compelling requirements that make the houses resilient.

Er. **Rabin Chaulagain** mainly shed light on the loss and damage caused due to 2015 Gorkha earthquake and highlighted the reconstruction and recovery initiatives in post-earthquake situations with prime focus on Post Disaster Needs Assessment (PDNA). The loss and damage caused by Lamjung earthquake (2021), Khotang and Doti earthquake (2022) and the detailed damage assessment conducted in affected districts following the earthquake was also briefly explained by Er. Chaulagain. He also shared the key challenges faced in recovery and reconstruction after the earthquake. The major challenges he focused on were assessing damages and identifying the needs of affected people. Along with it the funding and resource mobilization and resettlement and relocation of affected people were on the top. Besides the lack of skilled technicians and proper technical assistance is always a great challenge in shelter construction. He also highlighted the importance of alternative and local resources in shelter construction and reconstruction. Regards to post-earthquake reconstruction and recovery,



Er. Chaulagain also highlighted about the standard socio-technical assistance guidance for POs which includes Mason Training Curriculum, Engineer's Training Curriculum, Social Mobilizers Training Curriculum, Training of Trainers (ToT), Housing Design Catalog, IEC Materials and Standard Technical Manuals and guidelines.

The sharing done by Mr. **Khagendra Awasthi** elaborated on enhancing the disaster management process in Nepal through practice of good governance mechanisms in response and recovery initiatives. The Nepal Constitution 2015 and the DRRM Act 2074 serves as the main foundation principles for Nepal's disaster governance architecture. According to the constitution, the DRRM is both the sole authority of local government and a shared authority between the three tiers of government. He also highlighted the nine minimum characteristics for community

resilience. The DRRM and the Constitution of Nepal serve as the foundation for its disaster response policies. According to the Constitution, local governments must handle disasters first, and if they are unable to do so, the province and federal governments must step in.

Key takeaways of the presentations by four speakers largely underlined how to build resilience in the face of multiple hazards and climate change impacts. The major policies relevant to DRRM such as Disaster Reduction National Strategic Plan of action 2018-2030 and National Disaster Resilience Framework (NDRF) 2030, SRSP Guideline (Draft) and its significant implementation at local, district and provincial level are the guiding principles to achieve the key action



plan of disaster management ie. disaster preparedness, disaster response, disaster reconstruction and rehabilitation and disaster mitigation.



YouTube Link: <https://youtu.be/d4c3X7zBM2Q>

Presentation: <https://bit.ly/400DYRm>

7.2 Learning from post-earthquake reconstruction: How localization worked and what were the gaps?

The parallel session aimed to disseminate the lessons learnt out of the program and research initiatives implementation initiatives from 2015 Nepal earthquake recovery as well as the mechanisms of Build Back Better (BBB) concept applied during reconstruction and recovery of private houses. The speakers and their presentation focused on:

- Learnings from Post-Earthquake Housing Reconstruction, Mr. Aashish Tiwari, Structural Engineer, NSET
- 'Baliyo' Build Back Better: An inquiry into the Choice of Construction Materials for Housing Reconstruction in Gyalchowk, Gorkha, Mr. Tobias Vokuhl, Researcher, Oxford Brookes University
- Stakeholders Coordination for Localizing Seismic Resilience, Mr. Krishna Prasad Dhital, Chairperson, Rural Awareness and Development Organization (RADO) Nepal
- Nepal Disaster Resilience Localization process for Reconstruction and Settlement Recovery, Technical Coordinator - Mr. Rijan Gajurel, NHSRP

Session Chair/Moderator: Mr. **Kosh Nath Adhikari**, Project Director, CLPIU - Building

Facilitator: Ms. **Reshma Shrestha**, CRS

Er. **Ashish Tiwari** outlined the challenges and issues during the post-earthquake reconstruction phase including designing housing models, difficult geographical terrain, lack of sufficient human resources, complex urban issues, and the reluctance of banks to provide soft loans. The houses rebuilt soon after the earthquake by the affected people themselves and its non-compliance issues was one of the setbacks of NRA's reconstruction program. Er. Tiwari also presented the progress made in housing reconstruction, where 84% of beneficiaries received the third tranche by the end of 2021, which is considered successful compared to other countries earthquake reconstruction status like Pakistan and India. He also highlighted the Baliyo Ghar Project, supported by USAID, which focused on promoting



appropriate technologies, comprehensive approaches, and socio-technical support and contributed in Nepal's reconstruction largely through mason training, retrofit engineer training, retrofit model houses training etc. A survey conducted last year showed that 80% of masons who received training through the project are still working as professionals, indicating the effectiveness of the training provided by Baliyo Ghar Project.

Mr. **Tobias Vokuhl** presented on the pilot stage of his PhD fieldwork, and emerging themes from qualitative data collection on post-earthquake housing reconstruction modality in Gyalchowk, Gorkha district. His research is concerned with Build Back Better thinking in post disaster settings, and he demonstrated that holistically better houses and homes need to be not only 'baliyo' but need to meet many other context specific criteria for building users. He touched on the challenges to

householders in relation to making future proof housing decisions (owner driven reconstruction model) when countries like Nepal are affected by significant socio-political and socio-economic changes. Based on the researcher's early findings urban house forms are currently copy-pasted into village settings but may not be the 'better' house that will meet most livelihood, health, cultural, social, and religious needs in the best fashion. Further to the research, Mr. Vokuhl will collect additional data on usage of 'Build Back Better' approach from disaster response and resilient reconstruction.

Dr. **Krishna Prasad Dhital** from RADO Nepal presented on the significance of stakeholder's coordination for localizing seismic resilience where he highlighted the challenges of beneficiaries to follow the housing reconstruction design provided by the government and its limitation which leads to construction of small



houses. Mr. Dhital further shared the activities undertaken by RADO-Nepal, which followed an integrated approach for comprehensive capacity development to achieve safer construction in Makawanpur district. He also highlighted the major learnings from that period, connecting with the utmost necessity of stakeholder's joint efforts during the reconstruction and recovery phase for seismic resilience.

Mr. **Rijan Gajurel** from NHSRP delivered a brief update on the progress of reconstruction efforts in Nepal, covering areas such as private housing, heritage sites, government buildings and other public infrastructure too. He also discussed the modalities of grant disbursement by former NRA and NDRRMA, which shed light on clarifying the roles and responsibilities of these organizations. Mr. Gajurel briefly explained the Monsoon Preparedness and Recovery Plan of NDRRMA which is prepared on an annual basis. Mr. Gajurel through his sharing amplified that the current policies have not been able to address the people affected by the Doti earthquake. Decentralization of policies and responsibilities showered upon the local level smoothens the decision making which contributes in meeting the reconstruction efforts better targeted towards community needs. The presentation by Mr. Gajurel also showed the stakeholders mapping in Disaster Risk Management that contains

information of government stakeholders and active organizations working on that sector.

The four speakers in their presentations revisited the recovery and reconstruction efforts that were implemented after the Gorkha earthquake event. They emphasized the importance of engaging with local actors, such as communities, local organizations, and government entities, and highlighted how this approach contributed to the success of the reconstruction process. By involving local actors in the recovery and reconstruction efforts, community participation and ownership can be scaled up which were found instrumental to build overall resilience in the communities.



YouTube Link: <https://youtu.be/Yg0W5V6Yivw>

Presentation: <https://bit.ly/3Fk4aNg>

7.3 Multi-hazard DRR and climate change adaptation

The parallel session aimed to focus on documenting the several initiatives on responding to multi-hazard disaster recovery and initiatives on responding to climate change adaptation. The speakers and their presentation focused on:

- Use of Multi-hazard Scenario Ensembles to Inform Better Decision Making and Reduce the Impacts of Multi-hazards in Nepal, Ms. Sweata Sijapati, SAJAG Program-NSET
- Shelter and Housing Pre-Crisis Market Analysis, Mr. Rupendra Basnet, Habitat for Humanity
- Multi-Hazard Risk Informed City Planning, Ms. Bandana Sharma, USAID-Tayar
- Adaptation Solutions: Flood Resilient Habitat and Heat Stress Management in Urban Housing, Ms. Erica Udas, Sustainable Human Settlement, ICIMOD

Session Chair/Moderator: Mr. Krishna Gaire, Under Secretary, NDRRMA

Facilitator: Mr. Rabin Chaulagain, NHSRP

Dr. **Sweata Sijapati** through her sharing elaborated the ongoing research activities of Sajag-Nepal project, a multi-disciplinary research project including practitioners,



government, and academics from Nepal, the UK, Canada, and New Zealand with an aim to explore local knowledge and experiences of mountain hazards, including earthquakes and landslides, and their associated risk, and opportunities for more effective preparedness and response planning. In the context of emergency preparedness and response plan, only a single hazard is considered and the cascading impact of disasters such as earthquakes, monsoon which often results in landslides, debris runoff is neglected thus indicating the need of a multi-hazard plan in Nepal. Regarding this, Sajag-Nepal till date has conducted 3 rounds of Focused Group Discussions (FGDs) with the humanitarian clusters. The first round of the FGD gave an overview of how, who and when are the decisions made, the information used to prepare the contingency planning and what is missing. The second round helped to identify the most relevant spatial and temporal resolutions for understanding

monsoon-related impacts where most of the clusters preferred higher spatial resolution (building-scale) for the 72-hour forecasts. The third FGD focused on sharing the first results from the earthquake multi-hazard scenario ensemble which used the ensemble of 30 individual earthquake scenarios that underpin the current earthquake ERPP. For each modelled earthquake, impacts on buildings due to shaking damage were combined with impacts due to landslide occurrence and debris runout. The multi-hazard scenario ensemble clearly showed that with the consideration of multi-hazard there is the shift in the impact area. As the research is still ongoing, the future work would include inclusion of the monsoon outlook in the multi-hazard scenario ensembles and risk model along with incorporation of the 2021 census data to identify caseloads and estimate the size of the exposed and vulnerable population.

Mr. **Rupendra Basnet** and Ms. **Prakriti Nepal** representing Habitat for Humanity Nepal shared key findings on the assessment of Shelter and Housing Pre-Crisis Market Analysis (SPCMA) from the Far Western, Eastern and Karnali region. The findings of the assessment were that the construction market is quite volatile with the shortage of construction materials that need to be imported compared to locally available materials. The assessment revealed that the warehouse for construction materials at the local level is necessary for effective disaster



response. The Conditional Cash Transfer Program was designed and implemented to support families in purchasing construction materials and hiring workers also. HfH is an active member of the Cash Working Group, a shared platform for different organizations working in cash transfers through the banking system. Through several initiatives of HfH, cash assistance was also provided during COVID through the Assistance Management Committee composed of local women, youths, community users etc.





Ms. **Bandana Sharma** from USAID Tayar Nepal Project shed lights on risk informed land use planning and its relevance in multi-hazard city planning by working with eight partner municipalities of Nepal. The presentation recalled the documentation of impact studies of the 2015 Nepal earthquake, the subsequent flood and landslide and its long-term effect in the housing sector. The study found that due to lack of a multi-hazard risk assessment approach, the poor and vulnerable people must bear the compounded effects of disaster and climate change. Based on this background scenario, USAID DAI Tayar Project supported 8 municipalities namely - Bhimeshwor, Neelakantha, Rajapur, Lamkichuha, Godawari, Dullu, Birendranagar Municipalities and Tulsipur Sub-Metropolitan City in developing Risk Sensitive Land Use Planning (RSLUP) that minimizes risks for safer settlements and helps planned urbanization. The key features of RSLUP includes the

identification of disaster risk to aid urban development through land use tools and participatory approach. RSLUP promotes safer settlements and creates resilient communities. The RSLUP process includes various stages of working with municipalities in DRR and climate change adaptation interventions that include multi-hazard risk analysis, risk sensitive urban plans, land use policy, planning regulations and building bylaws, and risk sensitive land use plan implementation and monitoring plans to promote safer settlements. This plan is under implementation approval process by the municipalities.

Ms. **Erica Udas** shared key highlights of the pilot program implemented by ICIMOD on adaptation solutions that mainly covered the flood resilient habitat and heat stress management in urban housing. Owing to the effects of climate change, fluctuating scenario of monsoon and cascading impacts of disasters, a lot of them has affected Hindu Kush Himalayan Region which spreads across eight countries from Afghanistan in the west to Myanmar in the east. Adaptation and resilience building solutions have been identified that are GESI responsive, replicable, scalable, and affordable to all including diverse stakeholders and the communities. Pilot 1 was related to adaptation solutions on flood resilient housing in India and Bangladesh with raised foundation housing incorporating livelihood programs embedded in it such as homestead gardening, poultry farming

along with a sanitation program having raised plinth eco-san toilets. Pilot 2 included modified blue panel roofs for low-income houses in India to replace cemented or corrugated metal sheets. The water resistant and modular blue panels roofs are developed from recycled materials and have heat stress relieving and thermal stability properties that provide cooling effect to the inhabitants.

The examples provided by the speakers highlighted various strategies and approaches to address hazards, risk management, and climate change adaptation. These examples include use of geo-technical information sources of local geology to support planners and decision makers to develop appropriate mitigation measures. Similarly, use of pre-market analysis tools to assess potential risks, helps decision-makers to identify and address potential risks in the early stages of planning. Likewise, adaptation solutions and interventions should be



tailored to local contexts, considering the geography, socioeconomic and cultural settings, potential hazards/risks resulting in vulnerability of the area. Along with this, embedding data driven approaches help to identify vulnerabilities of socio-ecological systems, and design adaptation solutions including preparedness for potential hazards/risks and uncertainties of climate change impacts.

YouTube Link: <https://youtu.be/IqdTlF7B328>

Presentation: <https://bit.ly/3QpegD6>

7.4 To the Core of the Localization: Working with people, communities, and local government

The parallel session aimed to discuss the localization initiatives related to disaster affected private housing programs, safer and affordable housing practices, and municipal cooperation with partners on DRRM. The speakers and their presentation focus on:

- Municipal Led Safe and Affordable Housing Program, Mr. Raj Kumar Das, Chief Administrative Officer, Shambhunath Municipality
- DRRM localization Training: Lessons Learned and Way Forward, Ms. Ruchi Thapa, Project Lead, IOM
- People/Community Centered Approaches, Mr. Manish Raj Timsina, Shelter Coordinator, NRCS/IFRC
- Municipal Collaboration with Partners- Experience working with CityNet, Mr. Bishnu Paudel, Planning Officer, Tansen Municipality

Session Chair/Moderator: Mr. Santosh Kumar Khatri, Section Officer, MoFAGA

Facilitator: Ms. Reshma Shrestha, CRS

Mr. Rajkumar Das, representing

Shambhunath Municipality (Saptari district) highlighted the precarious living conditions of the marginalized flood affected families and elaborated the mechanisms adopted by Palika to support them with affordable shelters and assured their rights over healthcare. The initiative focused on partnering with families to build homes using local practices and materials, with bamboo being a key element due to its structural safety and treated nature. Basic engineering training was provided to enhance community skills. The project also promoted basic literacy and empowered the communities which helped them to interact with the visitors coming for research and exploring the learning. Mr. Das also discussed the positive impact of affordable housing on the community and shared the story of local beneficiary, Kampa Devi who regained her health and security in her new home supported through the localized approach of Palika. Overall, the presentation showcased the commitment



of local leadership to address the housing needs of helpless and vulnerable families while promoting sustainable development.



Mr. **Bishnu Paudel** representing CityNet National Chapter (CNNC) shed light on the municipal collaboration with partners, specifically the experience of working with CityNet -an Asia Pacific alliance of cities, and CNNC in Nepal. Mr. Paudel provided an overview of CityNet, its establishment, and its commitment to sustainable development of cities, towns, and emerging cities in the Asia Pacific region, similarly, emphasizing their efforts in advancing urban issues, promoting exchange of knowledge and best practices. It also showcases the activities undertaken by CNNC in Nepal, including coordination with the CityNet Secretariat, national meetings, capacity development programs, small project implementation, participation in webinars and publications. The overall focus is on sustainable development, addressing urban

needs, advocacy, networking and adapting impactful solutions to the local context.

Mr. **Manish Timsina**, IFRC-NRCS focused on the people and community centered approach in shelter reconstruction and rehabilitation during disaster response. The key messages emphasized that shelter response went beyond material distribution, considering localized approaches for understanding people's needs and the duration of their stay. Building back safer was crucial for successful interventions. Key lessons and the way forward included the effectiveness of a community-driven approach, flexible layout design, timely construction, use of local resources, integration of response initiatives into recovery actions, and the importance of beneficiary communication and collaboration with government authorities. The presentation also emphasized the need for integrating shelter with WASH, health including protection, gender



and inclusion initiatives and involving sectoral engagements and ensuring their ownership in the communities to meet quality benchmarks. Overall, the presentation promoted a holistic approach to sheltering, focusing on community participation, sustainability, and meeting the humanitarian needs of beneficiaries.

Ms. **Ruchi Thapa** from IOM focused on the Disaster Risk Management Localization Training conducted in Nepal. The sharing provided sufficient background information, including the endorsement of the DRRM Act 2017 along with needs and capacity assessment studies conducted by IOM in collaboration with key ministries. Based on the study findings, a training module was developed, and pilot trainings were planned and conducted in selected municipalities. The DRRM Localization Training Manual was endorsed in 2020 by the MoFAGA. An overview of Manual highlighting its utmost importance in capacity building for municipal local heads



and frontliner staff was provided. The 2 -days training were completed in all 753 local bodies of Nepal that contributed to the development of local DRRM acts and policies, mainstreaming DRRM in local development plans and instilled understanding of financing and risk transfer options. IOM Nepal organized 23 Master training of trainers' program at both federal and provincial level and further cascaded the training at all 753 local levels. The massive training movement at the local level, was made possible through the joint consortium partnership with NSET, Lutheran World Federation and Practical Action Consulting. A total of 19,904 local level representatives including local government officials benefitted from the training out of which 4,871 are female participants. This localization training is a flagship program of the Government of Nepal in terms of building capacity at the local level.

The sharing by four presenters focused on the initiatives undertaken by the local government in Nepal to enhance DRR response capacity. The key highlight was the DRRM Manual Localization training conducted in 753 municipalities across Nepal. The presenters shared insights into the training activities and emphasized the significance of these efforts. The DRRM Manual Localization training aimed to empower local government officials and stakeholders with the knowledge and skills necessary to effectively respond to

disasters. By localizing the DRRM manual, the training ensured that it aligns with the specific context and needs of each municipality. This approach increased the relevance and applicability of the manual, enhancing its effectiveness as a practical resource.



YouTube Link: <https://youtu.be/f9W1xkIaaq0>
Presentation: <https://bit.ly/3QkR8W1>







8. Key Parallel Session - Day 2

8.1 Multi-Hazard Risk Assessment

The parallel session aimed to share the experiences, lessons learned and way forward for the program initiatives based on evidence-based research outcomes of multi-hazard disaster risk assessment. The speakers and their presentations focused on:

- Earthquake risk assessment: Experience using the Global Earthquake Model in Nepal (GEM), Mr. Surya Narayan Shrestha, Executive Director, NSET
- Multi-hazard risk assessment: Experience -Dr. Basanta Adhikari,

Institute of Engineering/Tribhuvan University

- Multi-hazard risk assessment for the road sector- Er. Jiwan Guragain, Resilient Transport Specialist, Asian Disaster Preparedness Centre
- Risk assessment from Lamjung District: CRS's experience- Dr. Uttam Paudel, Technical Advisor-DRR, CRS
- Using Bipad Portal to collate and communicate risk, hazard, exposure, and vulnerability information- Mr. Pradip Khatiwada, Executive Director, Youth Innovation Lab

Session Chair/Moderator: Mr. Anil Pokhrel,

Chief Executive, NDRRMA

Facilitator: Ms. Sanchita Neupane, Program Manager, PIN

Mr. **Surya Narayan Shrestha**, Executive Director of NSET talked about the earthquake risk assessment and experiences using the Global Earthquake Model in Nepal (GEM). Elaboration on the risk assessment concept through hazard, exposure, vulnerability of various natural calamities such as earthquake, landslide, flood, and others were also done by Mr. Shrestha. Global Earthquake Model (GEM) is a non-profit public- private non-governmental organization founded in 2009 in Italy that deals in developing software, tools, and data for usage in earthquake risk assessment worldwide. Overtime, this model has developed trust in the assessment process which fosters solid applications in DRR. GEM is focusing attention on multi-hazard landslides and flood assessment in Nepal.



Dr. **Basanta Raj Adhikari** from Institute of Engineering (IoE), Tribhuvan University talked in detail about the wide usage of multi-hazard risk assessment and usage of early warning systems in Nepal and its relevance in the current period to warn people about the possible risk. In fact, the early warning system has saved the lives of many at the top of the list, though there is countless loss of private and public properties. The establishment of hydro-meteorological stations at the high altitude of 23000 m has been helpful to understand the high mountain climate and mountain patterns. Dr. Adhikari further shared that hazard and its associated risk factor changes at different pace over time and therefore calls for assessing the dynamic risk landscape and its cascading effects.



Er. **Jiwan Guragain** representing Asian Disaster Preparedness Centre (ADPC) informed the conference participants about the risk assessment project in the road sector of Nepal at Bagmati and Madhesh

Province, which is going to be completed soon. The study has used drone surveys and conducted detailed geological study of roadside areas. Er. Guragain also elaborated the scope of ADPC's working sector including of agriculture, water, transport, and policy, planning and finance in Nepal, Bangladesh, and Pakistan through CARE for South Asia Project, 5-year regional program (2020-2025) funded by World Bank that aims to promote evidence-based climate smart decision making and scaling up of laws, norms and capacities for climate resilient development. The project develops guidelines, capacity enhancement on revising standards and guidelines for transport infrastructures in Nepal. ADPC applies hazard, vulnerability, risk, and criticality assessment (HVRCA) framework in identified sectors and the detailed multi-hazard risk assessment.

Dr. **Uttam Paudel** and Er. **Anish Khanal** representing CRS elaborated the CRS' approach to DRR and risk assessment. They shared experiences from the Lamjung Recovery Project, Gorkha Landslide Response, and Geohazard Technical Solutions. Their approach involved localization of DRRM by contributing to systems strengthening for effective resource mobilization, risk assessment, visualization and communication for risk-informed actions, prioritization of actions and leveraging of resources for risk reduction. The team was able to support 13 local governments for geohazard



assessment of 29 sites across Gandaki and Lumbini Province, and assessments covered more than 500 households. The findings showed that due to the geo-physical and climatic settings of shelter and settlements in Nepal, many are under geohazard risk and require assessment and recommended mitigating actions. However, there is a lack of specialized human resources for such assessment's studies mainly at provincial, district, and local governments. So far, a



rapidly deployable team at the provincial level to support the municipal engineers, is critical. The province and national government are willing to allocate funds for mitigating actions if plans and requests are shared by the local government.

Mr. **Pradip Khatiwada** from Youth Innovation Lab, discussed the use of the BIPAD Portal for collating and communicating risk, hazard, exposure, and vulnerability information. The presentation also covers the importance of integrated disaster information management and the adaptation of open data and open-source technology for early warning and action. The use of Visualization of Risk (Viz Risk), a tool for better decision making through overlaying hazard, exposure, capacity, and vulnerability datasets, is highlighted. The presentation also emphasized the importance of engaging youth and young professionals in bridging the science-policy gap. Using a storytelling approach, complete disaster pictures can be communicated to support municipal governments in understanding when and where disasters may happen, plan for mitigation and preparedness, prioritize areas for emergency response, and assist in planning, budgeting, and implementation to reduce impacts and losses.

YouTube Link: <https://youtu.be/hJd3OFaUYZE>
Presentation: <https://bit.ly/3QjO4JW>



The presentations in this session mainly focused on the science-oriented techniques to multi-hazard risk assessment and highlighted the importance of understanding the dynamic nature of hazards and their associated risk factors over time. Community-level validation of risk maps and planning, individual household assessments, and geo-hazard assessments were among the tools used to identify risk reduction activities and development of localized solutions. Key takeaways from the session focused on assessing the dynamic risk landscape that involves continuous monitoring and analysing these changes to identify emerging risks and their potential cascading effects. It requires considering both direct and indirect consequences that can arise from a hazard event.

8.2 Integrating Gender, Disability and Social Inclusion (GEDSI)

The parallel session mainly aimed at highlighting the policies and programs influential in integrating GEDSI aspects in DRRM, as well as showcased the best practices in ensuring inclusiveness of women, people with disability, vulnerable populations etc in all possible phases of housing reconstruction and recovery. The speakers and their presentation focused on:

- Integrating GEDSI in DRRM; Sharing of Draft National Strategic Action Plan, Ms. Reena Chaudhary, GEDSI Focal Person, NDRRMA
- Building Resilience through Inclusive and Climate-Action Disaster Risk Reduction, Ms. Januka Neupane, Bagmati Province Coordinator, WHDRRP
- Durable Local Solutions Approach for Inclusion and Resilience, Ms. Arishma Shrestha, GESI and Protection Lead, PIN
- Integrated Approach for Shelter Housing and Settlements, Mr. Bhusan Aryal, Construction Specialist, Plan International

Session Chair/Moderator: Mr. Rudra Singh Tamang, Joint Secretary, NDRRMA

Facilitator: Mr. Rupendra Basnet, Habitat for Humanity



Ms. **Reena Chaudhary**, GEDSI Focal Person at NDRRMA emphasized on addressing the specific vulnerabilities of women and vulnerable groups of people through intersectional and gender-sensitive approach. This approach recognizes the diversity of vulnerable people's experiences and ensures that their voices are heard and included in decision-making processes by providing opportunities for women's participation in DRRM activities and ensuring that their specific needs are addressed in disaster response and recovery efforts.

Furthermore, Ms. Chaudhary provided a concise overview of some of the international frameworks such as Beijing Declaration and Platform for Action (BPfA), Paris Agreement 2008, Sendai Framework 2015-2030 to name a few. She further explained about the laws at the national level too citing the Constitutional Provision whereby listing exclusive rights of local

GEDSI Strategic Action Plan

The formulation of a strategic action plan is necessary to ensure the inclusive and accessible participation of vulnerable groups in decision-making and disaster risk reduction action in Nepal.

NDRRMA is committed to promote shock and gender responsive social protection, inclusive DRR efforts following the concept of leave no one behind to address the specific needs, diversity and vulnerabilities of most marginalized groups, different gender, people with disability children and senior citizens etc. The lack of an integrated mechanism or tool to address the sensitivity of vulnerable groups in the DRRM sector highlighted the need to mainstream GEDSI in DRRM, therefore NDRRMA through Task Force Group has worked out to develop draft document on “GEDSI Strategic Action Plan”. The document is expected to raise awareness and bring necessary changes in attitudes and behaviours, provide accessibility, strengthen resilience, and increase meaningful participation in national DRR endeavours.

The strategic action plan aims to bridge the gaps identified within the most vulnerable groups in all four priority areas of the Sendai Framework (2015-2030). It also considers the eight priority areas of the proposed National Disaster Resilience Framework (NDRF 2021-2030) and aims to reduce disaster risks and losses through the identification and formulation of strategic activities to be implemented in the short, medium, and long terms.

levels in disaster management mainly and the policies that were coordinated by NDRRMA. Nepal’s Constitution, adopted in 2015, recognizes the importance of disaster risk reduction and management, and includes several provisions related to disaster management at the local level.

Dr. **Januka Neupane** provided an overview of the Women Humanitarian and DRR Platform (WHDRRP) as a coalition of women humanitarian and DRR professionals in Nepal. Currently, the three provinces have established



committees for women's humanitarian aid, disaster risk reduction and management, and are considering expanding to the remaining four provinces. Dr. Neupane stated that it is crucial to ensure inclusivity in the Disaster Management Committee to ensure the safety of marginalized communities. Effective policies need to be developed and implemented in all stages of climate change mitigation and disaster risk reduction with mandatory participation of women. This requires consistent advocacy efforts. There is also

a need to increase the capacity of women in the workforce, including those engaged in the working group, through coordinated efforts. Managing the information system effectively and improving communication and coordination between the working committee and government and partner organizations at all levels is crucial.

Ms. **Arishma Shrestha** from PIN presented on the Durable Local Solutions Approach for Inclusion and Resilience. PIN combined localized solutions with science-based

Durable solutions: localized solutions

Durable local solutions are key to providing sustainable solutions to the challenges faced by communities affected by disasters such as the Nepal earthquake of 2015. PIN has been working closely with local communities in hard-to-reach areas to find solutions that address the specific needs of these communities. This approach has involved infrastructure rehabilitation and relocation solutions, such as housing, land and property, which have been implemented alongside science-based methods and advocacy.

PIN has been facilitating the implementation of the former NRA's resettlement program for households at risk of or displaced by geo-hazards after the earthquake. This has involved joint land ownership to increase access to assets and resources, and the inclusion of landless households in the relocation and rehabilitation procedure. PIN has also been implementing housing programs for the most vulnerable household's post-earthquake and providing access to vital documents and government social security allowances.

As an integral part of durable solutions, PIN has been working to increase the resilience of communities before, during, and after disasters by developing replicable solutions in disaster preparedness, anticipatory action, and emergency response by building long-term partnerships with government entities, CSOs, and the private sector, and engaging in evidence-based advocacy for more inclusive policies and longer-term climate-smart decision-making.

methods and advocacy. This approach highlighted the importance of facilitating and implementing durable solutions programs. Ms. Shrestha mentioned that using this facilitation approach, PIN successfully unlocked governmental grants worth USD 43 million to help the last mile population, including the displaced and landless population, relocate and rebuild after the earthquake. It was learned that PIN's next steps involve addressing the disconnection between climate change and disaster risk reduction at all levels in Nepal through a locally led intersectoral approach. They are co-producing an interactive tool called the "Climate Change Toy Model," which will help local governments and communities identify long-term climate change impacts and make investments in sustainable and resilient settlements and communities, such as multi-hazard climate-resilient houses for disaster-prone communities. Throughout all these efforts, PIN remains committed to ensuring that no one is left behind.

Mr. **Bhusan Aryal** from Plan International Nepal emphasized the importance of adopting a holistic approach to meet the needs of individuals in emergency situations, especially in promoting children's rights and gender equality. While ensuring access to safe shelter is a fundamental requirement, an integrated approach to shelter and settlement requires a comprehensive understanding of the needs of affected populations, the provision



of emergency shelter and integration with other basic services, attention to GEDSI, community engagement, and longer-term solutions that link humanitarian response to development goals. Mr. Aryal also drew attention of conference participants about Plan International's humanitarian-oriented interventions from a child-centred, girl-centred approach as well as using gender-transformative approaches to ensure that the needs of the most vulnerable are



addressed. The organization also prioritizes advocacy, child safeguarding and child protection standards in disaster risk management to ensure that children and young people (youths) are protected from harm during emergencies.

The session underscored the significance of an inclusive approach where GEDSI is at the core of effective practices which cater to the needs of local communities. Importance of cultural sensitivity, language accessibility, and understanding of local contexts in

disaster management is equally important where first responders and households must grasp the local level in response. Increasing awareness on this issue and working towards inclusive solutions with communities is essential to make this a priority in government programs. Importance of promoting self-sufficiency in local communities and the inclusion of GEDSI across sectors were urged as a common voice by the participants.

YouTube Link: <https://youtu.be/Zqd2jfWdpFo>
Presentation: <https://bit.ly/3ZZnRDy>

8.3 Greening Shelter and Sustainable Housing Practices

The parallel session aimed to share about the several proven and tested options for greening the shelter and resilient practices for sustainable housing. The speakers and their presentation focused on:

- Greening The Shelter Response, Ms. Preeti Karki and Mr. Laxmi Prasad Bhatta, NHSRP
- Bamboo Housing for Climate Resilience and Sustainable Living, Mr. Pranab Pradhan, Green Bamboo Creation Private Ltd.
- Low-cost Climate Friendly Houses through Micro Enterprises, Mr. Bjorn

Soderberg, Build Up Nepal

- Approaches for Energy and Resource Efficient Buildings, Ms. Dibya Acharya, Sr. Architect, MinErgy
- Environment Country Profile, Mr. Ajay Kumar KC, Water Sanitation and Environment Specialist- Consultant, NRCS/IFRC

Session Chair/Moderator: Mr. Dhruba Bahadur Khadka, Under Secretary, NDRRMA

Facilitator: Ms. Preeti Karki, NHSRP

The presentation by Ms. **Preeti Karki** and Mr. **Laxmi Prasad Bhatta** from NHSRP, concerned over the environmental impacts of the 2015 Nepal earthquake, valued the need of 'greening the shelter response project'. The presenter duo shared activities



conducted by the project and its impacts, and analysis in the overall response and reconstruction sector. Multi-hazard disasters mainly affect the housing sector, hence, the greening the shelter project focuses on recycling and reuse of the resources and the materials, adopting energy efficiency technologies, designing water and sanitation interventions, environment impact assessments and capacity building for recovery and reconstruction. In the absence of any green building codes and by laws, the duo emphasized on the green building guidelines, its institutionalization and dissemination. So far, the project has analyzed 10 policies of Nepal from a green perspective, identified its gaps and made recommendations. Therefore, the project aims to make an environment checklist and tool development such as sustainability scorecard approach to address the identified gaps.

Mr. **Pranab Pradhan** from Green Bamboo



Creation Pvt. Ltd shed light on the advantages of bamboo as green gold, its socio-economic viability, and the impacts and outcomes of using bamboo in resilient housing practices. Speaker raised the importance of bamboo as the green gold and highly renewable product, the best alternative to wood, plastic, and metals. Since bamboo is the fastest growing plant, its use can be very efficient in constructing houses and can also be used as fuel without



harming the environment. The successful construction work using bamboo in Pharping and Sundarijal were also shown by the speaker that showcased two different types of bamboo housings, one with the use of cement for the base known as Cement bamboo frame technology (CBFT)/Baroque technology and the other with just bamboo. The durability and viability of bamboo housings has a warranty of 30 years which in fact depends upon the species of bamboo used. Bamboo is proven to be used for constructing resilient environmentally friendly houses in the regions with high risk of flood, however, the approval from Government of Nepal on taking bamboo as construction materials has been a challenging journey for organizations and companies working in bamboo.

Mr. **Bjorn Soderberg** from Build Up Nepal spoke about different housing construction mechanisms using blocks prepared with the use of soil and sand or stone powder which has proven as eco-friendly as it reduces 75% carbon dioxide (CO₂) emission and is 25% less costly in comparison to the traditional brick housing. Some specialties of Compressed Stabilized Earthen Block (CSEB) bricks were elaborated that ensure the strength of the walls, are cheaper by 35-50% and are thinner with earthquake resistant quality. The initiatives on CSEB bricks is empowering while they work on a scalable micro enterprise model that teaches this method of construction to the local people in the community which



induces employment opportunities in local communities. 306 entrepreneurs have been positively impacted till date and 86,387 tons of CO₂ emission have been saved while constructing 8656 resilient houses.

Ms. **Dibya Acharya** from MinErgy centered her sharing about the project initiated by MinErgy towards achieving a clean, green, and energy-efficient Nepal. Ms. Acharya tried to bring to spotlight the necessity of energy efficiency in buildings



with examples of housing constructed by the initiation in the west of Nepal. The presenter talked about active and passive housing construction that impacts the efficiency of energy use in buildings. The team of MinErgy has been working on construction of such houses in different parts of Nepal to develop resilient housing and settlement practices. Four provinces of different bio-climatic regions have already been covered till date and the team aims to spread the practice all over Nepal.

Mr. **Ajay Kumar KC** representing IFRC-NRCS enlightened the participants by sharing the Environment Country Profile of Nepal for shelter and settlements program. Speaker focused on the research that aims to show our Country Profile in context to environmental impact and the need to minimize the impact of response activities and mitigate risk. The presentation provided relevant country level environmental information and identified the environmental component of housing and settlement. The data for development of the profile were taken from the database of government, census 2021, reports, scientific research papers and journals. This has attempted to inform the humanitarian programs in achieving sustainability, encourage resource efficiency and improve ecological values.



The parallel session emphasized takeaways that focused on promoting environmental-friendly practices and incorporating energy efficiency and waste management into recovery and resilient housing practices. The participants emphasized the importance of considering the ecological and cultural context of Nepal and integrating indigenous practices that are durable and sustainable. One of the key suggestions was to prioritize greener and more innovative approaches for post-disaster housing and settlements. This would involve exploring sustainable construction materials, incorporating renewable energy sources, and implementing waste management systems that minimize environmental impact.

YouTube Link: <https://youtu.be/IprG7STHuP0>
Presentation: <https://bit.ly/48UsuTr>

Greening the Shelter Response

Shelter and settlement responses carry one of the highest risks of negative environmental impacts in a humanitarian response, including risks of increased deforestation or overuse of natural resources, and the frequent use of plastic materials which contribute significantly to greenhouse gas emissions (GHG). To tackle this issue, the Global Shelter Cluster (GSC) has strengthened its focus to advance and support partners to reduce negative environmental impacts of shelter responses and consider climate change adaptation and mitigation measures.

“Greening the Shelter Response” being implemented by NHSRP aligning with GSC’s visions, has been designed to contribute to the greening efforts of GSC. It aims to enhance local capacities to facilitate a greener and climate-smart response by providing information, best examples, and other resources to greening the shelter response and reduce the emissions of greenhouse gases. The localization efforts in this initiative targets to support local actors in system & capacity strengthening through building upon existing knowledge, developing expertise, and the sustainable use of local materials. The project through its activities aims to pave the pathway towards greener humanitarian shelter and settlements responses.

8.4 Affordable Housing Solutions with Innovative Practices

The parallel session aimed to share experiences and document the initiatives from affordable housing solutions that were applied through innovative practices. The speakers and their presentations focused on:

- Green and affordable building materials for housing construction, Mr. Shuva Sharma and Dr. Santosh Shrestha, Scott Wilson Nepal
- Affordable Housing Solutions, Mr. Suraj

Shrestha, DRRM Coordinator, CARE Nepal

- Affordable Housing Finance Solutions to Address Housing needs of Low-Income Communities, Ms. Lajana Manandhar, Executive Director, Lumanti
- Lime Stabilized Soil Initiative-Field Experience from Banke District, Mr. Rupesh Shrestha, Technical Coordinator, CRS Nepal
- Shelter Solution through Housing Microfinance, Mr. Sunil Khanal, Sr. Manager, HfH

Session Chair/Moderator: Mr. Dhruba Bahadur Khadka, Under Secretary,

NDRRMA

Facilitator: Mr. Laxmi Prasad Bhatta, NHSRP

Dr. **Santosh Shrestha** and Mr. **Shuva Sharma** representing Scott Wilson Nepal provided an overview of its interventions on environmental initiatives as well as the alternatives for green structures and materials in private housing reconstruction and recovery. Both jointly put forward particular emphasis on proven, locally available, and viable environmentally friendly solutions that are widely accessible at the local level. Scott Wilson Nepal demonstrated the energy-efficient models in their interventions and disclosed that collaboration with the government to formulate the Green Building Guideline is ongoing, which is expected to serve as the primary document for guiding building code during the reconstruction and rehabilitation process following the disaster.



Mr. **Suraj Shrestha** from CARE Nepal gave a description of the CSEB (Compressed Stabilized Earth Brick) technology, a cost-effective housing option compared to conventional construction materials for the disaster-affected communities. Mr. Shrestha also informed the conference participants that CARE Nepal has already adopted this technology in Kailali as pilot program and is expected as good approach to boost the local economy by utilizing



locally available resources and mobilizing local saving and lending group. There is opportunity for women entrepreneurs and community groups to initiate this business, which would also help to create jobs for the area with small investment. There is still effort need to support on certification process of CSEBs for endorsement of construction norms of CSEB and making it eligible for public procurement process.

Ms. **Lajana Manandhar** from Lumanti Support Group for Shelter discussed home financing availability for housing the poor communities. An overview on how low-income households may receive housing financing and government initiatives to facilitate financial access was shared at the beginning of the presentation. Glaring gap in government policy has been seen which remains as one of the biggest obstacles for low-income families to access the suitable and inexpensive land. Lumanti was one of the pioneering NGO working for the urban poor people, forwarding the help to women's saving groups at the local level for accessing flexible financing through cooperatives for various purposes such as livelihood, health and primarily for land and housing. By using this strategy, women's cooperatives which were supported by Lumanti, have been able to have a collective savings of USD 17 million, which can be used as a source of financing for affordable housing for these low-income households. The private sector/ commercial bank was similarly brought in



by Lumanti to offer subsidized loans for the integrated settlement in Pokhara. Nearly 1109 households coordinated by Lumanti were able to find safe and affordable housing because of this strategy.

Ar. **Rupesh Shrestha** from CRS Nepal provided information on the pilot project conducted in Banke district to improve resilience to fire and flooding affected people. Mr. Shrestha informed that there are tests being carried out at the Banke



district for the Lime Stabilized Soil (LSS) Pilot project. During the fire and flood catastrophe, the use of lime stabilized soil as a binding element for building construction could serve as strong resistance. Due to the unique ability of lime stabilized soil to remain undissolved in water and its low carbon emission level, the LSS may be an excellent alternative for building construction in areas with a high risk of fire and flooding. To scale up the results of the pilot, CRS is assisting Bheri Technical School in offering capacity-building training on the use of lime as a building material.

Mr. **Sunil Khanal** from HfH discussed the key findings of the study on shelter solutions from housing microfinance. The study shows that the microfinance institutions are emerging as one of the most important resources for low-income households in Nepal who face difficulty to get housing solutions because large financial organizations primarily offer home loans to high salaried individuals. Most low-income families rely heavily on remittances as a source of income; thus, the study discovered that microfinance institutions are easier to work with when applying for a home loan. The study also revealed that there should be a supportive environment for low-income families to apply for subsidized loans for housing and that local authorities might play a significant role in informing low-income families about the availability of such access to finance. It became evident that



the government lacked the competence to consider policy in terms of providing low-income communities with accessible financial options.

The key takeaway from the session was the importance of affordable housing in the context of post-disaster recovery in Nepal. The participants extensively discussed the availability of accessible tools and technologies, as well as financing options, for housing reconstruction and rehabilitation after disasters. The session emphasized the need to explore alternative construction materials that are more disaster-resistant than traditional concrete materials. By adopting such materials, communities can enhance the resilience of their homes and reduce the risk of future damage. Organizations like Lumanti and Habitat for Humanity (HfH) were highlighted for their efforts in assisting low-income communities in accessing affordable housing funding options.

These organizations also provide viable technological solutions for post-disaster reconstruction, ensuring that the housing solutions are both feasible and sustainable.

YouTube Link: <https://youtu.be/a-bH0Xx6PqE>
Presentation: <https://bit.ly/45AuJZl>

8.5 Integrating WASH, Livelihoods, Land for Building Resilient Homes and Communities

The parallel session aimed to showcase experiences, best practices, lessons learned and a way forward from integrating WASH, livelihoods, and land related programs for scaling up the effectiveness in the building of resilient homes and communities of the disaster affected people. The speakers and their presentation focused on:

- Integration of Water Scheme, Livelihood and Shelter Component for Building Resilient Homes and Communities, Mr. Mehul Savla, Program Manager, CRS Nepal
- Indigenous Knowledge and Practice, Mr. Gajendra Thapaliya, Chief Administrative Officer, Jhapa Rural Municipality
- Innovative and Green Housing Practices, Ms. Salina Pradhan, Architect, HfH
- The impacts of Earthquake Disaster on Educational Accessibility, Affordability

and Continuity in Nepal, Mr. Nirmal Chongbang, PhD Student, Tribhuvan University

Session Chair: Mr. Chandra Bahadur Shiwakoti, Under Secretary, MoLMCPA (Ministry of Land Management Cooperatives and Poverty Alleviation)

Moderator: Mr. Kamala KC, British Embassy (UKAid) Kathmandu

Mr. **Mehul Savla** representing CRS elaborated the achievements and activities under the RECOVER Project which is a 3-years project that works with 6240 households across five districts spread across Gandaki and Bagmati province.



Mr. **Manish Maharjan** from ReCoVER's partner, Lumanti Support Group for Shelter also shared key highlights from the partnership initiatives. Integrated approach was implemented for safe reconstruction and improved economic resilience and livelihood opportunities of the targeted households among which financial literacy training, income generation support, market linkages, CASH support, input support and rehabilitation of WASH facilities were implemented. Three models of shelter support were implemented that included - top-up financial support to complete reconstruction of Rs 50,000, customized hands-on support to complete reconstruction or retrofitting and full construction support for 300 families. Identifying the most vulnerable families across five districts remained a big challenge during the early phase of the project for which local governments were coordinated. Continuous social-technical assistance (STA) were proved to



be supportive and helpful throughout the project implementation phase. Similarly, integration of shelter with WASH and livelihood were found crucial and a way forward for sustainability.

Mr. **Gajendra Thapaliya** from Jhapa Rural Municipality shed light on the use of indigenous knowledge and practice for rehabilitation of WASH and livelihoods for the disaster vulnerable populations. 40% households live in substandard houses,



most belong to indigenous groups in Jhapa Rural Municipality. Monsoon induced flood of 2017 destroyed many houses. As response initiatives, temporary shelters were created using locally available materials like bamboo. Using bamboo for constructing permanent homes was also practiced utilizing bamboo. In fact, the local communities were trained to utilize bamboo as construction material for temporary shelter by themselves. 918 houses for flood affected people have been built in the

last five years. Jhapa Rural Municipality allocated Rs 15 million annually for housing program support prioritizing the indigenous background of the communities for reconstruction intervention. Similarly, tenure security was equally prioritized for working with local land commissions for land allocation to landless families and managed to provide joint land titles to new families for the reconstruction.

Ms. **Salina Pradhan** from Habitat for Humanity Nepal shared the beneficial aspects of bamboo as the light weight, quick and easy to assemble, locally available and culturally accepted construction material. Beneficiaries in the working areas of HfH were trained and capacitated along with the local masons on how to work with bamboo. As a result, affected communities themselves were able to construct their temporary shelters themselves using bamboo. During construction, RCC structures were used for foundations and pillars while the walls were made using treated bamboo and mud. Ms. Pradhan also discussed the treatment of bamboo which in fact increases its durability, reduces insect infestation, and improves fungal infestation. Through HfH's programs, bamboo was rampantly promoted in construction because of its cost-effectiveness, cultural acceptance, and locally available green materials. The local economy is also supported as locals work as masons. Despite the successes of bamboo usage, bamboo is yet not validated by the



Government of Nepal as a construction material, it's not included in National Building Code and Guidelines. But many countries around the globe have building codes for structural bamboo. There is a possibility to develop one for Nepal as well in partnership with the local government for promoting green and affordable construction materials.

Mr. **Nirmal Chongbang**, a doctoral degree student at Tribhuvan University shared the synopsis of disaster impacts on educational accessibility, affordability and continuity in Nepal as part of the Ph.D research. Qualitative research in seven municipalities of Gorkha were conducted in which 214 households were surveyed and discussed post the Gorkha earthquake to understand the impact of the earthquake on students and their education continuity. Key findings of the research study outlines - 38% respondents said children's education was affected by the earthquake; 39%

respondents said they were unable to send children to school during the monsoon season; 38.78% respondents said that alternative livelihood opportunities were limited, while 4.67% respondents said that children without parents at home were not able to continue education. Recommendations from the study done by Mr. Chongbang suggests that a package of livelihood upliftment, skills transfer, local saving, and credit along with support on rehabilitation of sanitation points are the recurring needs of the communities. Similarly, the consulted people in the research area point to the crucial need of the comprehensive social protection package for the landless and marginalized families.

The session highlighted several key takeaways regarding the importance of a holistic approach to housing and the role of local governments in implementing housing programs. Some of the key points are – intersections of WASH, housing and livelihoods, local governments as housing implementation leaders, safe



land, green and affordable materials for housing, organizing settlements and tenure security and connecting livelihoods of disaster affected people. Overall, the holistic and sustainable approach led by local governments is crucial for achieving housing for all and resolving the issues faced by marginalized populations. Advocacy efforts and coordination with relevant authorities are necessary to bring about changes that prioritize local materials and sustainable construction practices.

YouTube Link: <https://youtu.be/m3zfK6CEtOI>

Presentation: <https://bit.ly/3tliHQc>

8.6 Multi Hazard Early Warning System

The parallel session aimed at sharing the importance of early warning systems, its services and usability in forecasting the probable disasters in Nepal such as in floods, landslides etc. On top of this, the session emphasized impact-based forecasting for anticipatory action in Nepal and lessons from joint action between NDRRMA, Department of Hydrology and Meteorology (DHM) and NRCS for data driven decision making. The speakers and their presentation focused on:

- Mr. Bikram Shrestha Juwa, Senior Divisional Hydrologist, DHM
- Forest Fire Early Warning System in Nepal, Mr. Sundar Sharma, Under Secretary (Tech.), NDRRMA
- Multi-hazard warning system: Nepal's progress and experience, Dr. Dharam Raj Uprety, Thematic Head of Climate & Resilience, Practical Action Nepal
- Mr. Manish Shrestha, Hydrologist, Water and Air, International Centre for Integrated Mountain Development (ICIMOD)
- Impact Based Forecasting for Anticipatory Action: Lessons from joint action between NDRRMA, DHM, Municipalities and NRCS, DRC, IFRC- Ms. Neeru Pradhan, Senior Program Officer, Danish Red Cross / Nepal Red

Cross Society

Session Chair: Rameshwar Marhatta, Joint Secretary, NDRRMA

Moderator: Mr. Sumit Dugar, British Embassy (UKAID) Kathmandu

Mr. **Bikram Shrestha Zoowa** from Department of Hydrology and Meteorology (DHM)/GoN mainly elaborated the importance of Early Warning System (EWS) and its services in Nepal. The traditional way of EWS in Nepal included the message flow from radios, television, and other broadcasting channels. There are about 2417 EWS which are installed in most large river system areas, but now the work is ongoing in small rivers too. Flood related to EWS is useful for the vulnerable people to save their lives and properties and for early preparedness. The backbone of EWS and services are institutional setup, human resources, technical capacity, funding, and infrastructure. The speaker provided examples of how effective and timely EWS



have helped in saving the lives of local citizens in face of loss to public and private infrastructures. Some gaps discussed by speakers in EWS include the wet season land use and soil moisture information, snowmelt information, high resolution flood and weather forecasting data, skills required for coupling the models, limited number of human resources in this sector and to name a few.

Mr. **Sundar Prasad Sharma**, Under Secretary of NDRRMA mainly explained the forest fire monitoring, fire related EWS, status and way forward for the multi-hazard EWS in Nepal. Mr. Sharma highlighted that forest fire skyrockets during April as the peak season in Nepal and draws the attention of everybody that changing climate context is one of the reasons for accelerating forest fires in Nepal. The initiatives taken from NDRRMA is to update the general public and diverse stakeholders about the real time data and its monitoring through BIPAD Portal and DRR Portal as well which is open



and accessible to all concerned. Mr. Sharma informed that the Ministry of Forest and Environment/Department of Forest and Soil Conservation/ICIMOD are still in the testing phase of data channelization and are developing fire risk outlook which is expected to contribute to forest fire detection and monitoring in Nepal.

Dr. **Dharam Raj Uprety** representing Practical Action Nepal shared the experiences of the community centric end



to end multi-hazard early warning system and the progress that Nepal has achieved so far in the sector. Dr. Uprety elaborated that climate change has impacted mountains, people, health, environment, agriculture, natural calamities, and livelihoods at large. Loss and damages are increasing and compelled due to multiple hazards. For example, data from January's 2012-October 2023 total people deaths figured around 3905, livestock deaths of 5977, infrastructure damage of 18659 and

economic loss amounted for 20000. Dr. Uprety informed the conference participants that in 2019, about thirty-five lakhs early warning SMS were delivered among the people and in 2022 1.5 million SMS were sent to the public exposed to flood risk in Nepal. Dr. Uprety reiterated that EWS should be community centric, impact based, and multi-hazards based. Tailored risk communication is essential for reducing risk of multiple hazards in lives and livelihoods of people and concerned authorities including local governments should prioritise early warning action and should increase investment on EWS infrastructures.

Mr. **Manish Shrestha** from ICIMOD highlighted ICIMOD's tools in aiding multi-hazard warning systems for Nepal. The thematic focus of the sharing by Mr. Shrestha relied mainly on using the tool on agriculture and food security, water resources and hydro-climatic disasters, weather and climate and air quality. A crucial NASA-USAID partnership aims to increase the ability of key stakeholders to incorporate earth observation data and geospatial technologies, which will improve environmental management and resilience to climate change. Its major focus is on science and technology innovations/advances to solve development challenges. The program initiatives of ICIMOD on early warning systems included building capacity of people and institutions to integrate science and technology and foster



strategic collaboration across scales and sections to achieve sustainable solutions. Mr. Shrestha also presented drought monitoring and early warning which provide forecasts up to 9 months. On top of the sharing is ICIMOD's High-Impact Weather Assessment Toolkit (HIWAT) which is a numerical weather prediction model that works 54 hours. HIWAT monitors rainfall, temperature, lightning strike, wind, flash flood, hail damage and works from Palika level. Modeling tools and satellite data is used to monitor the air quality, dust, and trace gas too.

Ms. **Niru Pradhan** from Danish Red Cross/NRCS shared about the impact-based forecasting for anticipatory action in Nepal and lessons from joint action between NDRRMA, DHM, Local government authorities and Nepal Red Cross Society. She explained about the general trigger model for forecast-based action (FbA) where multiple forecasting systems are monitored

and reported. The forecast is overlaid with household level flood exposure data to determine the level of impact. Hence, making the household level targeting easy and carrying out Early Actions. Hazard forecasting is also done in the intervention area under this joint action along with the FbA-SRSP Project implemented in Bardiya and Kailali from June 2020- October 2022. Based on this the communities have identified Early actions which were tested/ piloted and implemented during 2021/2022 floods.

The speakers emphasized the importance of making EWS as a community-centric model and prioritized community engagement. The speaker provided examples of how timely and efficient EWS have played a vital role in saving lives during emergencies while minimizing the damage to both public and private infrastructure. By involving and empowering local citizens, EWS can effectively communicate early warning information and enhance preparedness



at the community level. Similarly, several science-based partnerships were also discussed that focus on fostering scientific and technological innovations and advancements to address various development challenges. By leveraging these advancements, stakeholders can strengthen their ability to anticipate and respond to potential hazards, thus mitigating the impact on vulnerable communities.

YouTube Link: https://youtu.be/CVEpZ_dVfhw

Presentation: <https://bit.ly/3PT3GT2>

From Recovery to Resilience: Implementing Nepal's Declaration on Green, Resilient and Inclusive Development Pathway

Lesson Learned & Way Forward for a Green, Multihazard Resilient and Inclusive Recovery



9. Key Panel Sessions

9.1 Lesson Learned & Way Forward for a Green, Multi-hazard Resilient and Inclusive Recovery

Session Facilitator: Mr. Surya Narayan Shrestha, Executive Director, NSET.

An interactive panel discussion emphasized on the lesson learned from the Nepal earthquake and the way forward for multi-hazard resilient and inclusive disaster recovery was the major spotlight of Day 1. The session aimed to share lessons learned, challenges and experiences of the earthquake disaster response and how these learnings can

be integrated in the government, federal, provincial, local bodies. The Panellists assembled have enormous experiences in post-earthquake reconstruction and currently hold responsibility for disaster affected recovery initiatives in concerned government agencies.

Panelist: Mr. **Bharat Aryal**, Director General, Department of Local Infrastructure (DoLI)

Mr. Aryal held enormous experiences of contributing to the former NRA for around 6 years and reminisces NRA's work with technical experts, specialists and nearly 3300 engineers that were hired to monitor and safeguard the build back better and resilient housing reconstruction. Lack of

elected local governments were the major drawback during that time, despite this, through deployed engineers in affected districts, masons were trained on-the-spot and during the end of NRA's exit, nearly 90% private housing reconstruction were accomplished. Interactions and capacity enhancement orientation were conducted for elected local bodies from 2017 onwards in a total of 32 districts, that brought hope with placement of local leaders, however the utilization of the knowledge and



practices to the extent of success or failure of reconstruction is a question. Though Nepal government has given responsibility of disaster preparedness, disaster risk and reduction to central and provincial governments, local governments are the first port of call when any disaster strikes and therefore are the most responsible agency to reach to the people. Very useful guidelines and directives were developed by the former NRA which is still stressed as valuable reference. Since NRA has

transitioned this responsibility to another entity, the current procedures are found to have been delayed but through continuous intervention by local governments in the process of awareness towards safer reconstruction and owner driven approach, the achievements are expected to have spellbound impacts in the lives of peoples.

Panelist: Mr. **Kosh Nath Adhikari**, Project Director, CLPIU – Building and Housing

Mr. Adhikari shared his experiences by recalling the involvements in active NRA phases. Belonging to the technical engineering field, Mr. Adhikari delved into the massive promotion of the principle of Build Back Better as the mandatory metaphor to safer reconstruction. The revision of National Building Code (1994) was done with the formation of NRA in the Nepal reconstruction process that paved the way for resilient construction. Integrated settlements in small pockets of land in rural communities were developed



after the geohazard assessment where the households were connected to improved livelihoods such as ecotourism, homestays etc. Self-help perma system and revolving grants were revitalized during Nepal earthquake reconstruction time, along with the breakthrough involvement of the community in the heritage site reconstruction. Mr. Adhikari summarized that some districts and local governments have received flagship success which needs to be scaled in other similar areas. The massive scale of private housing reconstruction in 14 most affected districts and in 18 moderately affected districts must be further built up and expand the knowledge and learnings on resilient reconstruction to all 77 districts. Stating the vulnerability of Nepal towards monsoon induced disasters and earthquake, Mr. Adhikari further expressed that concerned government agencies and involved partner organizations should undertake the programs to replicate the best practices of NRA's reconstruction program wherever needed.

Panelist: Mr. **Dilli Ram Luitel**, Project Director, CLPIU - Education

Mr. Luitel underscored the huge loss and damage caused by the 2015 Nepal earthquake in the educational sector with complete collapse and damage of government and community school buildings that are mostly constructed in slope landslide prone areas. Because the earthquake happened on Saturday 25th



April 2015, huge loss of lives of students was prevented as a symbol of god's advice towards the next big step of learning. Tentatively around 9 thousand school buildings and 55000 thousand classrooms were destroyed by the Nepal earthquake. Following the formation of NRA, CLPIU-Education was also formed under NRA's authority that looked mainly at the reconstruction and rehabilitation of school buildings. Reconstruction of educational sector was undertaken through four models mainly:

- School Management Committees were given responsibility to reconstruct schools with the government budget and 7500 schools were constructed through SMCs.
- 700 schools were reconstructed with support from NGO/INGOs.
- 10% of those schools were constructed through building contractors.

- A small number of schools were rebuilt through turn-key model (example of Kathmandu's Durbar High School with Chinese government investment)

Reconstruction of some schools within heritage sites have been achieved also through CLPIU Education's interventions where communities were mobilized. 150 schools were retrofitted in total that were achieved through community driven retrofitting models. Reconstruction of government owned campus buildings is also ongoing. Throughout these processes, incorporation of guidelines for resilient reconstruction with drawing prototypes, and other essential criteria required were made mandatory and strictly followed.

Panelist: Mr. **Krishna Gaire**, Under Secretary, NDRRMA

Mr. Gaire recalls his deployment in Mugu district/Karnali province during 2006/07 where a massive natural disaster hit Mugu. He was involved in relief and response interventions throughout Mugu as well as when a Tornado hit Bara district heavily. Mr. Gaire highlighted that the disaster response works by development organizations did not reach real affected people and duplication of responses were rampant. Contrary to this, organizations tend to neglect the reconstruction part in post disaster situations. In Rautahat flooding, some cases of deaths of daughters in Muslim communities were seen which we were unable to investigate due to



religious reasons. In the Bajura landslide, relocation of vulnerable communities was found challenging. In Terai plain, mostly the housing and other infrastructures too are vulnerable to flooding as there is old age practice of building houses in low lying areas that are prone to waterlogging in the rainy season. Contextually, several laws and acts related with resilient reconstruction are present in all three levels of government, however, the laws and activities associated with that are overlapping. Solid institutional strengthening of local level, mainly targeting the capacity enhancement of community-oriented program staff is necessary for achieving the prosperous development.

Concluding remarks from Session Facilitator Mr. Surya Narayan Shrestha

Mr. Shrestha emphasized that the lessons learned from post-earthquake reconstruction and rehabilitation in Nepal can be effectively implemented

in the pathways of GRID. The principles and strategies employed during the reconstruction phase can be applied to achieve effective disaster recovery and rehabilitation at the community level. It was highlighted that maintaining uniformity in disaster response is crucial, and duplication of activities should be avoided. Coordinated efforts and clear roles and responsibilities among different stakeholders are necessary to ensure an efficient and effective response. The importance of involving and empowering local governments in all phases of disaster management, from preparedness to response, recovery, and reconstruction were highlighted by panellist. Local governments should have ownership and play an active role in these processes. Additionally, raising awareness at the community level is essential to enhance resilience.

The need to replicate the successful

practices and approaches used during the previous NRA period in similar disaster contexts is discussed thoroughly among majority of development actors and stakeholders. Building on the lessons learned and applying successful strategies can improve future disaster response and recovery efforts. Lastly, the platforms such like NSHSF for their effectiveness in implementing post-disaster reconstruction learnings. Such platforms provide avenues for sharing knowledge, collaboration, and driving the way forward for reconstruction efforts in the most effective manner. Overall, the session highlighted the importance of incorporating lessons from past experiences, involving local governments, ensuring uniformity in response, and leveraging effective platforms to enhance disaster recovery and resilience at the community level.

YouTube Link: <https://youtu.be/m9Yimhr-pCA>

9.2 Lessons from 2015 Earthquake Recovery

Session Facilitator: Mr. Sushil Gyewali, Immediate Chief Executive Officer of Former NRA

This panel session was held with a major objective to share and openly discuss the lessons learnt from Nepal earthquake in 2015 and the way forward for multi-hazard resilient & inclusive recovery. The panel discussed the way we have been moving forward after the earthquake in 2015 in context to NDRRMA. Mr. Taranidhi Bhattarai and Mr. Kishor Thapa talked about their book 'Bahattar Saal ko Bhukampa', its major highlights, and their way forward for better disaster management practices. It further elucidated the efforts of agencies in the sector of recovery and resilience, the learnings till date and the challenges faced throughout to lead the way forwards better.

Panelist: Mr. **Taranidhi Bhattarai**, Professor, Department of Geology, Trichandra Campus, Kathmandu

Dr. Bhattarai shared about challenges that his team of experts and students experienced while orienting the people and making them understand that a disastrous earthquake in Nepal is yet to come and they are predicting it to occur in the west. He added that the people are not ready to comprehend the need of preparedness, which led them to rush in building new



houses in the same traditional ways just to receive a grant from the government which has posed Nepal at much risk. However, the promising scenario established by former NRA in post disaster reconstruction has entailed that the local capacities have to be enhanced for injecting integration of resilient reconstruction and disaster recovery in everyday lives of peoples and communities, whether it means sensitizing people on safer construction, safe site selection, application of building codes etc. The guidelines developed in post-earthquake reconstruction phases are still considered valuable reference for current entities too.

Panelist: Mr. **Kishor Thapa**, Urban Planning Expert at Nepal Institute for Urban and Regional Studies

Mr. Thapa highlighted the fact that human casualties have decreased in the past years due to the preparatory activities conducted by the several government agencies and



private organizations too which calls for efficiency of the team in sensitizing the people with information dissemination. Mr. Thapa added that the disaster recovery activities carried by NDRRMA has accelerated in recent times as it has contributed to post monsoon induced disaster reconstruction, several equipment needs for disaster response such as during fire incidents as well as for search and rescue, mobile emergency ambulances purchase etc. He drew attention to the interventions by the former NRA in 32 districts and reiterated that the coverage should be increased to 77 districts by the current entity NDRRMA. Mr. Thapa recalled the success of former NRA by stating that due to a focused responsibility of earthquake reconstruction only has made the work easier for NRA to intervene. Mr. Thapa suggested that the diverse network of stakeholders and local governments should internalize the core of understanding recovery and resilience,

rather than just relief and response only.

Panelist: Mr. **Surya Narayan Shrestha**, Executive Director, NSET

Mr. Shrestha winded up the discussion by saying that gaps have been seen regarding service delivery due to unclear roles of NDRRMA and the local level governments along with the level of awareness among the community people about the emergency challenge response and understanding on the disaster recovery. Several exemplary initiatives were milestones of NRA backed up from support of development partners and involvement of I/NGOs. Promotion of BBB in reconstruction of damaged infrastructures, deployment of social mobilizers, mobile masons, housing for the vulnerable people, heritage reconstruction, low cost housing including the focus shifted to improving the livelihood of poor and marginalized affected communities are notable work during former NRA phase, the essence of which has potentials to replicate



further. Governmental processes to access grants to people are lengthy in current times too. Mr. Shrestha also proudly praised the effectiveness of NGO Mobilization Guidelines that provided a way forward to the regularization and harmonization of donor agencies and private organizations' involvement. Mr. Shrestha underscored the need to separate an entity to work in the sector that collaborates with the organizations working in resilience and recover. He also identified the gaps of former NRA's in scale-scope differentiation, joint donor engagement that created misalignment with regular donors and the periodic ones and a confusion regarding the responsibility and the roles of other organizations.

Panelist: **Mr. Minar Thapa Magar**, National Coordinator, NHSRP

Mr. Thapa Magar shared the experiences through the perspective of a donor and a funding agency. He shared that the



earthquake relief procedures and housing recovery has been successful because of the aid and support from international agencies and humanitarian agencies. Mr. Thapa Magar highlighted that the problem was likely to set grounds when the government announced 3 lakhs per household as a relief fund to the affected families for rebuilding the house from owner driven approach. Initially the NGO's showed a commitment in providing humanitarian services to the public, but at the same time when a long-term support was needed only a few of those remained constant while the others slowly backed off or exited from mainstream reconstruction works. The focus of NGO for providing reliefs only is not a long-term approach, reiterated by Mr. Magar and further raised attention that the need of socio-economic approach for a long-lasting sustainable response is required that should be an integral part of the development program.

Concluding remarks by Session Facilitator Mr. Sushil Gyewali

Mr. Gyewali raised concerns over the remaining housing reconstruction challenges and the need for a targeted and serious focus to complete the delayed reconstruction caseloads of Nepal earthquake 2015. Sharing his enormous experiences from the successful reconstruction carried out by the former NRA, Mr. Gyewali highlighted several key points regarding the challenges in housing reconstruction and the need for

a focused approach. One major concern was the improper budget allocation of NDRRMA, which contributed to setbacks in achieving the desired outcomes in housing reconstruction of disaster affected people. Another issue raised was the transfer of human resources and their placement within the government sector. It was suggested that officials with relevant educational backgrounds and experience in disaster management or related fields should be deployed in NDRRMA, which contributes to ensure the implementation of resilient DRRM practices from federal to local level. To strengthen effective DRRM, a socio-economic approach that prioritizes the most vulnerable and marginalized households is highly recommended. Mr Gyewali reiterated that local bodies are identified as key entities that efficiently verify and ensure the accuracy of data from the field so that the real affected



people are not overlooked. As a takeaway from the discussion, it was concluded that a permanent working group for disaster preparedness and response should be established. This would ensure continuous efforts and a proactive approach towards addressing the challenges associated with disasters and building resilience in the long term.

YouTube Link: https://youtu.be/_W9StjzBnE

9.3 Provincial Government Pathway towards Resilience

Session Facilitator: Mr. Minar Thapa Magar
National Coordinator of NHSRP

The objective of this panel session revolves around hearing from the representatives working at the provincial ministerial sector about the coordination mechanisms, their understanding towards DRRM, division of the roles and responsibilities towards Palikas, their correspondence mechanisms with federal governments and to the diverse stakeholders.

The panel discussion commenced with a brief presentation by Mr. Thapa Magar, emphasizing the multifaceted nature of disasters. He shared insights on budget planning and the allocation of skilled human resources for DRRM at the provincial and local levels. He emphasized the crucial role of federal governments in fostering coordination with provincial governments to enhance the effectiveness of DRRM initiatives.

Mr. Thapa Magar presented data on private housing damage caused by various hazards in each province of Nepal. The figures revealed the extent of multi-hazard housing damage in seven provinces, ranging from 934 to 18,648 housing damages. Additionally, the update on monsoon and fire-induced disasters highlighted

the number of housing damages in each province, with figures ranging from 3 to 4,263. The allocated DRRM budget for FY 2079/080 in the seven provinces was provided, ranging from NPR 6.30 Crore to NPR 23 Crore. Furthermore, the provincial monsoon/fire recovery and reconstruction fund updates were shared, indicating the allocated and spent budgets in each province. Finally, it was noted that there are nine types of DRRM documents, with Koshi and Karnali Provinces having the highest number of documents (seven out of nine), followed by Bagmati Province with four documents. Gandaki and Karnali Provinces have three documents each, Sudur Paschim has two, and Madhesh has only one.

The following figures reveal the extent of multi-hazard housing damage in seven provinces:

- Koshi: 1,576
- Madhesh: 934
- Bagmati: 2,186
- Gandaki: 6,129
- Lumbini: 1,346
- Karnali: 2,215
- Sudurpaschim: 18,648

Similarly, monsoon and fire induced disaster affected household update as of April 2023 in seven provinces is as follows:

- Koshi: 1,308



- Madhesh: 3
- Bagmati: 2,328
- Gandaki: 4,263
- Lumbini: 1,949
- Karnali: 2,759
- Sudurpaschim: 3,410

Likewise, the DRRM Budget allocated in FY79/80 in the seven provinces are as follows:

- Koshi: NPR 10.80 Crore
- Madhesh: NPR 6.30 Crore
- Bagmati: NPR 19.28 Crore
- Gandaki: NPR 23 Crore

- Lumbini: NPR 10 Crore
- Karnali: NPR 7.23 Crore
- Sudurpaschim: NPR 22 Crore

Similarly, provincial monsoon/fire recovery and reconstruction fund update are as follows:

Province	Budget Allocated	Budget Spent
Koshi	280 L	153 L
Madhesh	-	-
Bagmati	931.4 L	931.4 L
Gandaki	1658.5 L	1658.5 L
Lumbini	1000 L	456 L
Karnali	-	-
Sudurpaschim	-	-

* L = Lakhs

There are nine types of DRRM documents as follows. Koshi and Karnali Provinces have seven documents out of nine followed by Bagmati with four documents, Gandaki and Karnali with three each, Sudur Paschim with two and Madhesh having only one.

After the presentation, Mr. Magar initiated the discussion by posing thought-provoking questions to Panelists from provinces regarding the annual damage of approximately 6,000 houses.

DRRM Documents	Provinces						
	Koshi	Madhesh	Bagmati	Gandaki	Lumbini	Karnali	Sudurpaschim
DRRM Act	✓	✓	✓	✗	✓	✗	✗
DRRM Regulation	✗	✗	✗	✓	✗	✗	✗
DRRM Policy	✓	✗	✓	✗	✗	✓	✗
Strategic Action Plan	✓	✗	✓	✗	✗	✓	✗
Relief Standard	✓	✗	✗	✗	✗	✓	✗
Fund Mobilization Procedure	✓	✗	✓	✓	✗	✓	✓
Recovery & Reconstruction Guideline	✗	✗	✗	✗	✗	✓	✗
Rescue & Relief Distribution	✓	✗	✗	✗	✓	✓	✗
Provincial Disaster Response Plan	✓	✗	✗	✓	✓	✓	✓

Panelist: Mr. **Dhurba Gaida**, Secretary of the Ministry of Internal Affairs and Law in Bagmati Province

Mr. Gaida on the inquiry over provincial vision, plan and policies regarding fundamental right of safe housing, he highlighted the constitutional provision that has established three levels of government. He mentioned that the provincial government takes care of 30% of the safe shelter needs after monsoon-induced disasters. He added that a total of 9 crore has already been sent for this purpose.

Mr. Gaida highlighted several challenges in due course of time such as difficulty in verifying whether the 30% relief grant that was sent has been utilized properly and if the construction has adhered to the building codes. He also pointed out the challenges in distributing relief items. As per their provision, an amount of 25,000 is added per person or family. However, there seems to be a discrepancy in the approach. Medical professionals argue that they lack facilities and training to ensure proper distribution. He emphasized the importance of coordination in addressing these challenges and moving forward successfully. Mr. Gaida stressed on effective communication and information sharing between all levels of government as a crucial part in addressing DRRM as well as shelter reconstruction and recovery initiatives. Up-to-date and accurate information sharing can lead to avoiding duplicates in resources



utilization and response and recovery works of shelter as well as in early warning systems, evacuation plans and emergency response protocols. Collaboration is vital in addressing shelter and geographical challenges, thus enabling all relevant stakeholders to take appropriate actions in a timely manner.

Panelist: Mr. **Umesh Basnet**, Secretary of the Ministry of Internal Affairs and Law in Koshi Province

Mr. Basnet shared the initiatives of Koshi Province with budget allocation in DRRM. Additionally, they have constructed 145 sample houses and have recently obtained approval for the provincial disaster risk reduction policy. They are also awaiting approval for the strategic action plan by the Council. Mr. Basnet discussed the challenges related to shelter and geographical location. He highlighted that many people reside in hazard-prone areas, particularly near rivers, where relocation



is difficult due to geographical constraints and limited available land space. Moreover, individuals are often reluctant to leave their ancestral lands and relocate to safer areas. To address these challenges, Mr. Basnet proposed the implementation of integrated settlement development as a potential solution. This approach would require standardized working guidelines across all levels of government, as currently, practices vary among different administrative levels. Additionally, due to limited resources, collaboration and partnership with various stakeholders is essential to effectively tackle these issues. Mr. Basnet also emphasized the importance of incorporating new technologies and learning from past experiences to enhance disaster risk reduction efforts. Moreover, he highlighted the need for investment in disaster risk reduction and suggested that the private sector can play a role in disaster risk financing. Lastly, Mr. Basnet stressed the significance of effective communication

and information sharing between all levels of government to ensure coordination and collaboration in addressing shelter and geographical challenges in hazard-prone areas.

Panelist: Mr. **Suran Prasad Sharma Paudel**, Under Secretary of the Ministry of Internal Affairs and Law in Gandaki Province

As a part of provincial vision, plans and policies regarding right to safe housing and DRRM, Mr. Sharma mentioned provincial warehouses, the implementation of disaster management regulations, with 7 crores allocated specifically for disaster management and on top, allocation of 10 crores for private housing. Mr. Sharma highlighted several challenges for the future in his response. He identified three types of geographical structures in the region, each associated with different types of hazards such as glacial lake outburst floods (GLOF), floods, and landslides. He stressed that settlements located near



rivers are particularly at risk. One of the challenges mentioned is the lack of caution and awareness at the lower level. Despite the existence of laws and policies at different levels of government, there is still a lack of awareness regarding DRR. Therefore, raising awareness becomes crucial.

Another challenge is the lack of skilled human resources and expertise in the DRR sector. Government employees are often assigned as focal persons for a certain period before being transferred, resulting in a lack of long-term dedicated disaster experts at the local level. To address this, a team of professionals with expertise in disaster management should be established at the local level. The issue of limited resources and finance was also highlighted. Local levels often have limited income sources to cover their expenses, and the additional expenses required for DRR pose an extra burden. Furthermore, in the event of a major disaster, the capacity of the local level to handle it is often insufficient. Therefore, collaboration and cooperation among all three tiers of government are necessary to overcome these challenges effectively.

Panelist: Mr. **Nidhi Raj Neupane**, Under Secretary of the Ministry of Internal Affairs and Law in Karnali Province

Mr. Neupane discussed monsoon preparedness and other measures as part of provincial vision, plans and policies in DRRM. He mentioned the mobilization



of clusters and a Provincial Emergency Operation Centre (PEOC). He also emphasized the significance of the DRRM Budget, which handles all types of disasters. In their province, Humla, Jumla, Mugu, and Kalikot were severely affected, with over 8,000 houses damaged. Mr. Neupane estimated that an additional budget would be required, apart from the funds received from the central government. He also highlighted several crucial issues pertaining to disaster management. One of the key concerns he raised was the need for enhanced coordination and collaboration among different organizations and levels of government. This lack of coherence poses challenges to the overall effectiveness of disaster response. Mr. Neupane raised concerns that disaster management is not solely the responsibility of the Ministry of Internal Affairs and Law at provincial level, but also involves various other ministries responsible for health, infrastructure, and more. Thus, it is imperative for all

ministries to actively participate in disaster management, as the burden cannot be shouldered by a single entity. He emphasized the significance of both bottom-up and top-down coordination, as well as vertical and horizontal coordination across different levels and sectors.

Another significant challenge highlighted by Mr. Neupane is the shortage of skilled human resources at the provincial level. Although there has been some positive progress in terms of officials being required to stay in one place for a minimum duration of one year, the province still lacks an adequate number of competent personnel to effectively carry out disaster management tasks. Financial constraints were also identified as a major obstacle to development and disaster management efforts in Karnali Province. Mr. Neupane stressed the need for substantial financial resources, potentially in the billions, and called upon development partners to implement programs in Karnali region. He mentioned that despite having a budget allocated, the province has struggled to utilize it due to shortage of capable human resources. He also cited incomplete status of risk mapping works in the province due to scarcity of skilled human resources.

Panelist: Mr. **Dharma Raj Ojha**, Senior Land Survey Officer, Ministry of Land Management, Agriculture, and Cooperative, Sudur Paschim Province

Mr. Ojha highlighted the progress made



in disaster management and mentioned that 77 houses were destroyed in the Freed Kamaiya Camp fire incident. However, due to budget limitations, they faced challenges in addressing the situation. In this regard, Habitat for Humanity has provided support for the construction of 77 houses. Mr. Ojha further discussed specific areas prone to floods. The Raji basti area, for instance, is frequently affected by floods. To mitigate the risks, 25 houses have been constructed and provided to the Raji community. Similarly, Freed Kamaiya's concerns have also been addressed. Another example he mentioned was Dipayal, where the Badi community resides. The municipality has taken steps to address the risks they face. He also mentioned a significant flood incident in Surkhet in the year 2071, which resulted in encroachment and a substantial reduction in the width of the river from 100 meters to 20-30 meters. Additionally, he highlighted the flooding risk in Achham, particularly in the area next to

the Kailash River. For effective disaster risk reduction, he stressed if people should be allowed to build shelters in areas that are at risk. Secondly, he questioned the use of land and the mention of risk zones in the government maps and suggested the need for land use planning at all levels.

Mr. Ojha shed light on significant challenges related to securing land for building shelters. He emphasized that this is the foremost priority. However, he expressed concerns regarding the potential negative consequences of certain actions taken by local government bodies, such as the use of bulldozers, which can render existing residences unusable. To mitigate this issue, he proposed two potential solutions. The first suggestion is to acquire and provide land to individuals, ensuring they have suitable space for constructing residences. The second suggestion involves the National Land Commission allocating government-owned land specifically to landless Dalit communities, squatters, and unorganized settlements, before commencing the construction of shelters. These measures aim to address the critical need for secure land for housing purposes in Sudur Paschim Province.

He also raised concerns about unplanned development, specifically the construction of roads without proper land use planning which has led to debris flow from the hills to the Terai region, resulting in inundation problems. Proper land use planning is necessary to prevent such

issues and minimize the risk of disasters. Another problem he mentioned is the construction of houses in agricultural lands, particularly in the western belt of Kailali province. Houses are being built by creating bridges from the road. He urged serious consideration of whether such construction practices should be allowed. He emphasized the need for collaboration between those working on shelter projects and the National Land Commission to provide housing in risk-free areas, aiming to minimize the risk of disasters. In brief, Mr. Ojha emphasized the significance of thoughtful land management, proper land use planning, and risk-free housing allocation to mitigate the potential risks and challenges related to land and housing in Sudur Paschim Province.

Concluding Remarks by Session Chair/ Joint Secretary of MoHA, Mr. Pradip Koirala

Mr. Koirala shared concluding remarks by citing Nepal Constitution 2072 and important Annexes that mentioned disaster management in three instances - i) In Annex 7, both local and provincial levels have a common responsibility, ii) In Annex 9, all three levels - federal, provincial, and local - share the responsibility, iii) Local level holds the sole authority.

Mr. Koirala added that common responsibility becomes nobody's responsibility, it is essential to clearly define work responsibilities for each tier

of government based on the type and intensity of the disaster. This requires the development of comprehensive guidelines to guide the interventions actions of each level. Mr Koirala mentioned a 'Guideline on Delineation of Roles' that needed to be circulated in all provinces. Clarity in roles is crucial, such as allowing the local level to seek assistance from the provincial and central levels when they are unable to handle the situation themselves, and for the central and provincial levels to refrain from unnecessary interference and let the local level take charge. A decision from the Cabinet on these guidelines will greatly facilitate the process of DRRM. Mr. Koirala also emphasized the importance of understanding the sensitivities involved in DRRM and GEDSI aspects and requested everyone concerned for taking proper precautions on this.

Preparedness plays a vital role in reducing the need for extensive response efforts. Establishing a basket fund involving all stakeholders to support preparedness initiatives is necessary. The efficiency of provincial governments can be assessed



based on their professional handling of situations during the COVID-19 pandemic, from preparedness to responses mainly. Effective coordination requires information sharing, common reporting, joint decision-making, and joint consensus among three tiers of government to collectively determine the way forward. In conclusion, the main message from this national conference, in remembrance of the 2015 earthquake, is that successful DRRM during large-scale disasters is achievable through coordination and collective efforts in risk reduction.

YouTube Link: https://youtu.be/WfuB_s4kqX0
Presentation: <https://bit.ly/3QhULMt>

9.4 Local Government Pathway towards Resilience

Session Chair: Mr. Rudra Singh Tamang,
Joint Secretary, NDRRMA.

The panel session on local government pathways towards resilience was facilitated and led by Mr. Rijan Gajurel, Technical Coordinator, NHSRP. The objective of this session revolves around hearing from the representatives working in the local governance sector about the coordination mechanisms, their understanding towards DRRM, the roles and responsibilities of local bodies, their involvement in DRRM initiatives and the way forward for the federal governments and to the diverse stakeholders.

Panelist: Ms. **Laxmi Devi Pandey**,
Chairperson, National Association of Rural
Municipalities in Nepal (NARMIN)

Being the umbrella organization of entire rural municipalities of Nepal, NARMIN is involved in forwarding the issues, gaps and advocacy, joint coordination on relevant subject matters, capacity building programs, advocacy related initiatives, capacity building of elected members, support the Palikas as per immediate necessities whereby linking with federal and province level government. For NARMIN, coordination is done through factual evidence at all levels of government from local to provincial and federal.



NARMIN has managed to document the good practices in reducing risks that arise from disasters which are shared to all rural municipalities as the guiding document.

Ms. Pandey has answered the question pointing the active involvement of NARMIN in capacity building projects, advocacy, study of 460 municipalities on the status, capacity building of elected members, support of the Palikas as per immediate necessities by linking with federal and province level as well. NARMIN has Documented good practices in reducing risks that arise from disasters.

Chairperson of NARMIN, Ms. Pandey put emphasis on utilization of locally available resources and local skilled human resources so that environmental impacts are considered while carrying out reconstruction activities. Similarly, Ms. Pandey emphasized on one door policy and collaborative works by three tiers of government. Emergency Rescue Centre

is required to be established in all 753 local government bodies of Nepal with fulfillment of necessary equipment and skilled human resource with enough stockpiles.

Ms. Pandey further informed that NARMIN has managed to form a Rescue Team as a model with six members in Hupsekot Rural Municipality, Nawalpur district. The team is equipped with necessary skills and requested to replicate this similar practice in all 753 local government bodies of Nepal.

Panelist: **Mr. Jit Narayan Thapa Magar**, Chairperson, Budhiganga Rural Municipality, Morang district

Budhiganga Rural Municipality is extremely vulnerable as it shares a boundary with four rivers. The Palika prioritizes the need of preparedness activities including awareness focused programs, stockpiles, setting up of the ambulances, search, and rescue materials etc. to face the disaster. Contextually, coordination efforts are



highlighted in terms of joint work by assembling diverse stakeholders, networks, and communities at large. Mr. Thapa Magar recalls that the fragile areas of Budhiganga Rural Municipality are affected by floods every year with loss of lives and properties. Proper coordination with the federal and provincial level can resolve the issues of landless people who became landless as flood affected them. Mr. Shah drew an example of one well-off landlord who showed mercy and donated land to the landless people in Budhiganga areas. On the contrary, such brain-drain activities should be avoided and should have sought solutions of poverty and landlessness from the state itself as the state owns power and resources to solve the ground issues through proper dedication. Utilizing the platform of NSHSF, Mr. Thapa Magar requested funds and resources from the central government with emphasis put over for utilization of local resources to its maximum extent for disaster risk reduction.

Panelist: **Mr. Jay Narayan Shah**, Chairperson, Jhapa Rural Municipality, Jhapa district

According to Mr. Shah, DRRM at the local level is achieved solidly through the joint coordination works forwarded by all concerned agencies, therefore coordination is highlighted as an important instrument for preparedness and disaster response effectively. Mr. Shah pointed out that sufficient joint work has not been provided by the federal and provincial

level government to the local authorities. Mr. Shah recalled the devastating flood caused by the Kankai river. The areas of Jhapa Rural Municipality were hit hard, and many low-lying areas were inundated because of flood occurred on midnight of 17 August 2017. The flood affected nearly 80% land and almost all roads and bridges of Jhapa Municipality were inundated by flood. 12 people lost their lives while many people were rescued and more than thousand people became homeless and displaced by the ill-fated flood. The public of Jhapa Rural Municipality garnered a great lesson from the devastating flood that instead of running away in fearful state, it is wise to face the disasters and prepare for it in advance.

Chairperson Shah further elaborates that in the post flood situation, the Palika coordinated with HfH to rebuild the houses where a sum of NPR 150,000 was provided by HfH for the reconstruction of each new house utilizing the local resources such as bamboo, sand & concrete blocks. Due to the coordinated efforts, currently 40% of vulnerable families have been transferred to new homes that benefited around 800 vulnerable families whereas the remaining 60% will be transferred this year. Risk of disasters are reducible if there is advance planning and coordinated properly with like-minded development organizations such as HfH along with the utilization of local resources which has multiplier effects in the community itself with capacitated



local masons, job creation and improved livelihoods.

Panelist: Mr. **Kala Nidhi Devkota**, Executive Director, Municipal Association of Nepal (MuAN)

MuAN represents entire urban municipalities (including metropolis and sub-metropolitan cities) of Nepal under one umbrella organization. Mr. Devkota highlighted three different areas where



the local governments can practice benefit sharing utilizing the sole rights and in capacity enhancement areas. Firstly, prepare exemplary laws (policies) and send them to the local level to prepare local laws accordingly for smooth operation at the implementation level. Secondly, focus should be in the sector of capacity and resource management jointly by adjoining municipalities, for which Mr. Devkota cited an example of 11 municipalities in Belgium working together for jointly owned waste processing centers, where one compensates for the lack of resources, while another one looks after other deficit areas. Mr. Devkota raised the issue of managing fire trucks by single Palika which is quite expensive. Multiple Palikas can coordinate together for funding and utilizing the single fire truck. Mr. Devkota reiterated for delineation of significant coordination, work division and resources management among 3 tiers of government. Currently 25% of the fund is allocated to the local government to manage disasters, earlier it was 5%. The joint forces from diverse stakeholders should focus on inter municipality coordination for sound DRRM.

Panelist: Mr. **Bishnu Poudel**, General Secretary, CityNet National Chapter

CityNet is the largest association of urban stakeholders working for sustainable development in the Asia Pacific region. Established in 1987, the network of cities through CityNet has grown to include 171 municipalities, NGOs, private companies,



and research centers too. Citing the clause 225 of Constitution of Nepal that insists for coordination and coexistence, Mr. Poudel emphasized on decentralized approach, data informed decision making, division of resources and good coordination among diverse sectors to tackle varieties of local issues such as disaster and climate change. Mr. Poudel stated the joint policy discussion involving three tiers of government in the context of revision and update of policy that favors local governments.

Mr. Poudel further shed light on interventions of CityNet by taking examples of Bangladesh, India, and Indonesia where CityNet has prioritized some focused areas particularly in disaster and climate change. CityNet is about networking alliance for knowledge exchange, skill transfer from international to national level and vice versa. Earlier disaster means rescue, response and providing relief materials, but the modality has been changed to the

perspectives of preparedness and moving towards resilience. Mr. Poudel reiterated that all the local levels should maintain the capacity of becoming resourceful to handle the disasters, converge together to support in a time of need, exchange ideas and capacities. Suggestions for the one-door policy during disaster response execution by supporting agencies and following the mandate of government is the area where all local levels can move forward.

**Concluding Remarks by Session Chair/
Joint Secretary of NDRRMA Mr. Rudra
Singh Tamang**

Mr. Tamang emphasized the importance of effective coordination among the three tiers of government (local, provincial/state, and federal), to ensure a cohesive and well-coordinated response to disasters. Additionally, he highlighted the significance of inter-municipal coordination, where neighboring municipalities work together to address common challenges and share resources. He stressed for clear guidelines and adequate resources for coordination with local governments for effective planning and implementation of DRRM initiatives at provincial and local level mainly.



Mr. Tamang also stressed the need for the federal government to play a supportive role in disaster management. This implies that the central government should provide assistance, guidance, and resources to local governments and their representatives. Furthermore, he emphasized the localization of disaster management activities, primarily the disaster preparedness, response, and recovery efforts should be tailored to the specific needs and context of local communities. This approach recognizes that local governments and communities have a better understanding of their own vulnerabilities and can effectively address them through targeted actions.

YouTube Link: <https://youtu.be/cZT7HktAmPs>



10. Closing Session

10.1 Appreciation and Token of Love distribution

Council Member of NDRRMA, Dr. Ganga Lal Tuladhar handed over the Certificate of Appreciation to representatives of NSHSF Organizing Committee members and thanked them for their wonderful contribution in organizing NSHSF. Similarly, Joint Secretary of NDRRMA, Mr. Pradip Koirala handed over the green themed Token of Love gift green themed Token of Love gift distribution to the Panellist members, special guests, session chairs and chief guest.









10.2 Closing Remarks by NSHSF Secretariat and Organizing Committee

On behalf of NSHSF Organizing Committee Members and the Secretariat itself, Mr. **Rijan Gajurel** thanked everyone for the visible participation in the parallel and panel discussions as well as during the several networking sessions over tea and coffee and market stall. Further, Mr. Gajurel stated that the incredible achievements of Nepal in post-earthquake housing reconstruction and rehabilitation is commendable which is evident through

the vibrant sharing in the first national conference of NSHSF. Through the first national conference on NSHSF, not only have we managed to discuss the best practices, lessons learned but also have focused on the shortcomings, challenges, and way forward. This conference has opened avenues for many of us in like-minded thematic development areas, which has been possible through contributions by many. Organizing this kind of national event annually by a single organization is an uphill task but if we jointly coordinate and collaborate then everything is possible.



10.3 Closing Remarks from Council Member of NDRRMA, Dr. Ganga Lal Tuladhar

Council Member of NDRRMA, Dr. Ganga Lal Tuladhar shared the closing remarks as the guest. He mentioned that the reconstruction after the 2072 Baishak earthquake is itself a disaster risk reduction campaign and Dr. Tuladhar praised the commendable sharing in the two-days national event of NSHSF. The destruction we faced, and the huge challenges showered on us by the 2015 Nepal earthquake is evidence ahead of us that together we can work out many things. The post-earthquake reconstruction work is a good example of turning shortfalls into passion and power.

Dr. Tuladhar reiterated that the experience of 32 districts should be taken and applied in the rest of all other districts of Nepal for further effectiveness in disaster recovery. He said that the NDRRMA should be empowered more to lead the campaign of moving from recovery to resilience. The Government of Nepal has not only rebuilt the homes but also has expanded the cash grant mechanism through owner driven approach and established a robust multi-hazard housing recovery process throughout the nation. In fact, events like this act as a vibrant campaign in DRRM to review, reflect, plan, and proceed for the future. Nonetheless, organizing wonderful sharing forums like NSHSF it shows that together we can gain success in achieving Nepal as a resilient nation by 2030.



10.4 Closing Remarks from Chief Executive of NDRRMA, Mr. Anil Pokhrel

Speaking at the closing session, the Chief Executive of NDRRMA, Mr. Anil Pokhrel praised the active involvement of all concerned speakers, presenters, panelist, participants, and the organizing committee members. He further highlighted that first edition of national conference on NSHSF were envisioned to adhere by the theme of 'From Recovery to Resilience: Implementing Nepal's Declaration on Green, Resilient and

Inclusive Development Pathway". Scanning through the magnitude of sharing best practices in a two-days event, abundant learning has been instilled among many of us which is in fact a revolutionary example and a promising way forward in the field of achieving Nepal's declaration on GRID pathways. We would continue to work together synergistically and together in DRRM. We need cooperation and collaboration at all levels, from all agencies, communities and to promote the whole of society approach for enabling us to continue moving ahead into the GRID pathway and implementing it to the core of achieving resilience.



11. Participants Feedbacks and Reflection on NSHSF

Some vibrant sharing at two days of NSHSF wholeheartedly praised by participants were the local level led housing experience sharing, localized housing solutions, province and local governments' panel discussion about their housing priorities for safe housing, flood resilient housing solutions, increased interest in energy efficient technologies, affordable housings, green construction technology with a focus on easily accessible local products such like bamboo as a sustainable alternative, and the importance of an integrated approach including land for housing, WASH, and livelihoods improvements for resilient

homes and communities. In fact, the lessons learned, and best practices shared by organizations and researchers at large provided promising steps towards adding value to improving the disaster resilience of the homes and communities of Nepal (in context to the vulnerable geographic situation of Nepal).

Participants from various thematic cross cutting sectors of DRRM, post disaster reconstruction and resilience looked forward to witnessing a continuation of conversations through common platforms such as NSHSF. Participants observed that the discussions initiated at the

forum are beyond the normal discussions on DRRM, shelter reconstruction and settlements recovery and look forward to translation into partnerships, synergistic collaborations, and informed interventions.

On top of this, participants discussed that by involving local stakeholders in decision-making processes, policymakers can gain a better understanding of the specific context in which disasters occur. This understanding enables the development of targeted and effective policies that address the unique needs and challenges of each locality. Participants provided feedback over improving the timing and schedule of sessions to ensure sessions do not overlap and participants can attend all interested sessions. Ensuring gender balance, its

need and relevance were also reflected as feedback by participants and suggested for meaningful participation in upcoming days. Along with this, involving private sectors directly for tapping future fundings and collaboration was also suggested and streamlining of agenda considering the current scenario with in-depth discussions rather than covering a wide range of subjects in one basket.

A strategic advisory group consisting of government members, donors, organizing committee, and partners to provide strategic input and guidance was advised to form for upcoming days for enriched insights and adding value to the thrilling management.





12. Summary of Recommendations

- DRRM cannot be solely handled by the government alone, therefore, partner organizations should be actively involved through key collaborations, provided with a conducive platform and significant environment to contribute to development efforts in various cross-thematic sectors of DRRM and leverage external support in where able possible rescue, relief, response, reconstruction, and recovery can enhance the effectiveness of DRRM initiatives.
- To ensure effective coordination, the Provincial Emergency Operations Centre (PEOC) must remain active as DRRM is a shared responsibility, therefore the activeness of such entities will naturally be more visible due to responsibility takeover.
- Active participation from provincial governments is expected, particularly regarding the right-of-way for rivers and roads. Areas within these rights-of-way that are affected by disasters should not be eligible for relief.
- Risk-sensitive land use planning is crucial for identifying vulnerable areas. Categorizing zones into red (most vulnerable), orange (vulnerable), yellow

(less vulnerable), and green (safe) is important. People living in red and orange zones should not be made eligible for relief, but execution of necessary arrangements for relocation should be intervened. Local and provincial governments should coordinate effectively in making announcements regarding eligibility criteria of the areas that should be made public.

- The government should take proactive measures to address the issues faced by landless people and ensure their inclusion in the rehabilitation and reconstruction process with the 'Leave No One Behind' concept.
- One-door channelized reporting mechanism should be in place for relief and cash items received, ensuring monitoring, transparency, and accountability.
- Each community should have an open area capable of accommodating 20% of its population that cites the breakthrough roles that both local and provincial governments can play in contributing to effective DRRM.
- Strengthening coordination mechanisms and fostering better communication and collaboration between the central, provincial, and local levels of government is essential. This would help ensure a more effective and cohesive approach to DRRM.
- Enhance and increase the visible roles of provincial government in DRRM efforts by assigning a greater responsibility and authority to the provincial government in planning, implementing, and coordinating DRRM activities.
- Empowering the provincial government can strengthen leadership, improve better alignment and utilization of resources, expertise, and decision-making power at the provincial level to address DRRM challenges effectively.
- Deployment of dedicated and skilled government officials at the local and provincial level would facilitate better coordination, decision-making, and implementation of DRRM initiatives efficiently.
- Developing a stronger team, beyond just one or two staff members, is essential. These individuals should be trained in various aspects of DRRM, including awareness building, information collection, rescue operations, and relief distribution during times of disasters.
- Preparation of frontline community volunteers' squad (beyond just one or two skilled members) who can actively contribute to DRRM efforts should be prepared in each province and deployed at local level. These volunteers would play a crucial role in supporting and assisting communities during times of disaster.

- Frontline responders, such as the army and police, plays a crucial role in disaster response, therefore, capacity development efforts should focus on equipping these frontline personnels with the necessary skills and knowledge to effectively carry out their responsibilities during emergencies.



Annexes

Question and Answers (Q&A)

Q. What measures can be taken to enhance the effectiveness of accurate data & information sharing?

A. Collaborative efforts are adopted to engage with relevant stakeholders to ensure the accurate dissemination of information at the local level, that mitigates the negative implications associated with inaccurate information sharing. Necessary efforts are laid to develop an automated risk information system that aims to incorporate not only census data but also social vulnerability data to maximize the data efficiency in terms of time and effort.

Q. What approach can we adopt to utilize

scientific data at the local level, to reduce disaster risks and encourage decision-making based on data analysis?

A. There is a need to conduct in-depth studies and spend a significant amount of time gathering information and understanding the local context based on the received data.

Q. Why is the northern and western belt considered more prone to earthquake vulnerability than the Terai belt?

A. Scenario maps can sometimes be misleading and not always provide an accurate depiction. We are actively working on addressing these issues and will ensure that the results are shared with everyone.

Q. Recommendation process for tranche distribution has been halted since the past few months affecting the work of organizations facilitating reconstruction for vulnerable families. Since many families have not completed reconstruction, when will the recommendation process be resumed and what plans does CLPIU have to address this?

A. It is agreeable that the payment of tranches is yet to be completed and is being halted due to cabinet decision dated 26th Mangshir 2079. CLPIU will work towards settling this issue within this fiscal year.

Q. How risk sensitive land use planning (RSLUP) is being implemented in the working areas of Tayar Nepal Project?

A. Currently, there are ongoing discussions with municipalities to implement the RSLUP in the working areas of Tayar Nepal Project. In some project areas, the plan is being implemented partially, while in other municipalities, it is being fully implemented.

Q. What are the fire-resistant properties of the blue panels and which materials are used to make these panels.

A. The blue panels are composed of recycled packaging material and agricultural waste, while there is no conclusive guarantee regarding their flammability characteristics.

Q. How have the multi hazard caseloads been collected in the primary phase? What

kind of survey or methodologies followed to complete the stakeholders networking map and what kind of organizations are included?

A. The local government has been using RMIS & Kobo Collect Mobile Apps to collect data on the multi-hazard caseloads. Once the beneficiaries are verified by the district disaster coordination committee, they become eligible for grants of up to NPR 50,000 for temporary shelter. However, implementing this policy is challenging at all levels of government due to several gaps and issues.

Networking has been established through multiple meetings with provincial governments, and information has been collected through email and coordination. NHSRP will coordinate with other agencies active in disaster programming to ensure effective collaboration.

Q. What were the challenges or obstacles that caused delays in the process of urban recovery?

A. Due to the Nepal Rastra Bank's decision to limit access to subsidized loans exclusively to banks, low-income families are facing difficulties in applying for such loans due to collateral issues. Furthermore, the accountability of the private sector has been compromised because of the elimination of the Bank's mandated subsidized loan program, which offered loans at a low interest rate of 5%.

Q. Do organizations such as HfH and other entities involved in shelter initiatives consider the significance of family size, recognizing that houses of the same dimensions may not adequately accommodate a family of four compared to a family of eight?

A. During the design process, HfH and its involved partner organizations consult with community members and local government to ensure houses are suitable for a family of 4 to 5 members considering cultural practices where families may split after marriage. The house sizes are generally sufficient, considering budget constraints. However, provisions are made for future extensions through horizontal expansion, allowing homeowners to potentially add on to their houses if they choose to do so.

Q. Is there an alternative to using cement in construction during bamboo use? You mentioned that walls are built with bamboo and mud. How does this construction method ensure resilience against floods?

A. While mud is used in constructing walls, it does not imply that the houses are not flood resilient. Cement structures can also be susceptible to being washed away during major floods, as evidenced in past incidents. The concept of flood-resilient housing is somewhat contextual and varies based on the specific circumstances. In the lowlands where we operate, the floods typically

occur when rivers overflow during the monsoon season, causing certain low-lying areas to be submerged temporarily. To address this, we consider historical flood levels and elevate the plinth level accordingly. Additionally, our foundations and pillars are made of reinforced concrete (RCC) structures, providing stability and strength. The mud plaster is primarily utilized for the walls. Another important aspect for us is advocating for the allocation of safe land for new construction.

Q. What is the targeting approach employed in Project ReCoVER to ensure the inclusion of the most vulnerable individuals, considering the diverse range of disabilities and intersecting vulnerabilities?

A. Our local-level baseline survey offers comprehensive data regarding the specific vulnerabilities of the beneficiaries, enabling us to employ a robust selection process utilizing our most vulnerable scoring system. The criteria for participant selection are well-defined and consider the diverse range of disabilities and intersecting vulnerabilities.

Q. What is the expected lifespan of houses made from bamboo?

A. Treated bamboo typically lasts for around 30 years. However, with the use of cement bamboo technology, based on experiences from buildings in Colombia, the lifespan can be extended up to 100 years.

Q. What are the provincial visions, plans, and policies that ensure the fundamental right to safe housing in the face of annual damage to approximately 6,000 houses?

A. The provincial governments in Nepal have implemented various initiatives and measures to ensure the fundamental right to safe housing and address damages caused by hazards. These initiatives include budget allocations, construction of sample houses, disaster management regulations, establishment of emergency operation centers, and mobilization of clusters. The provincial governments are actively working towards disaster risk reduction and management, aiming to provide safe shelter and mitigate the impacts of disasters.

Q. How risk-sensitive land use planning is being addressed and implemented to mitigate potential hazards and ensure the safety of communities?

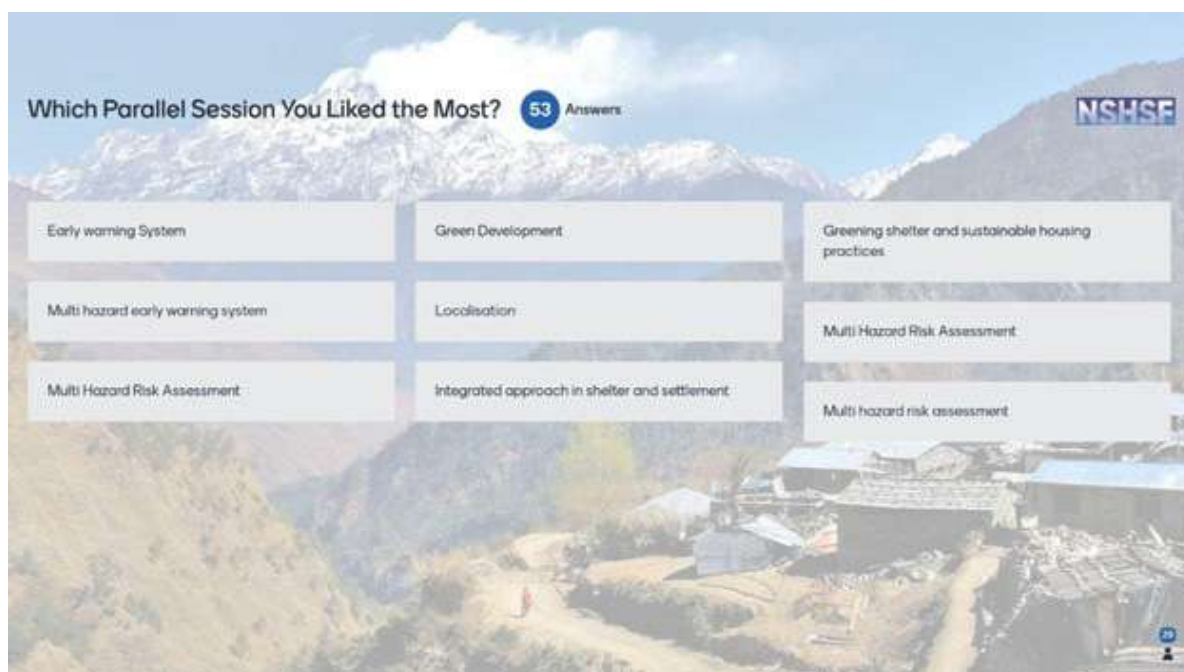
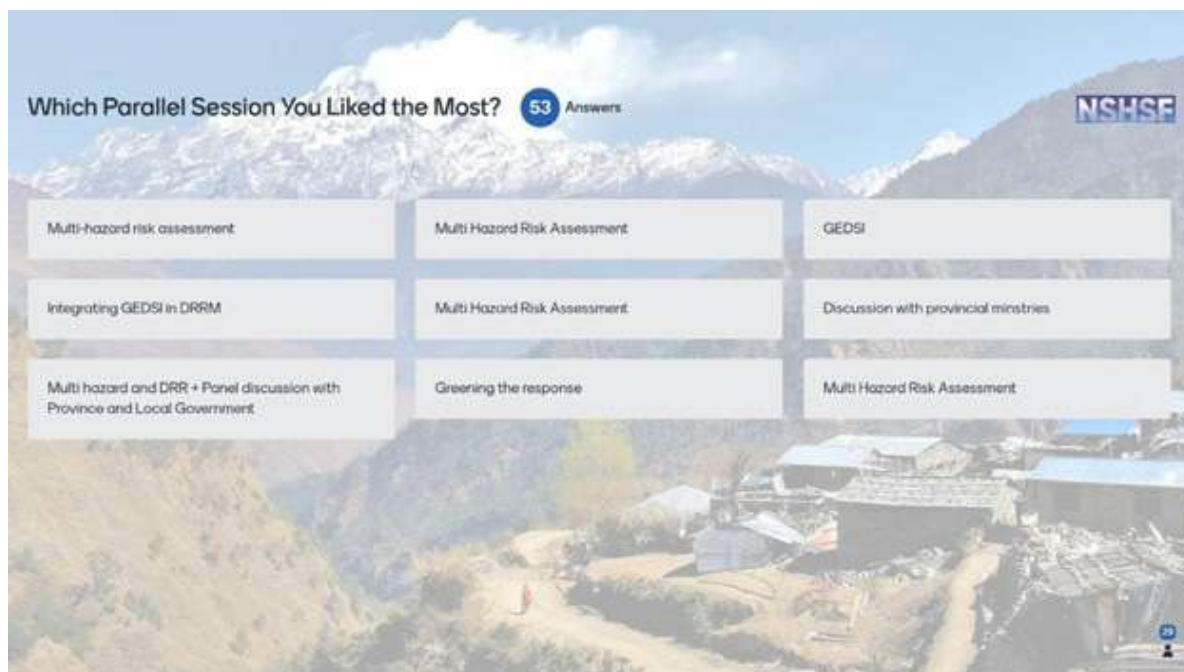
A. Risk-sensitive land use planning is being addressed and implemented in Sudur Paschim Province to mitigate potential hazards and ensure community safety. Efforts are being made to mitigate risks in flood-prone areas through the construction of houses for affected communities. The importance of accurate government maps depicting risk zones and the need for effective land use planning at all levels were emphasized. Collaboration with external organizations like Habitat for Humanity has played a role in addressing housing challenges caused by disasters. Land use

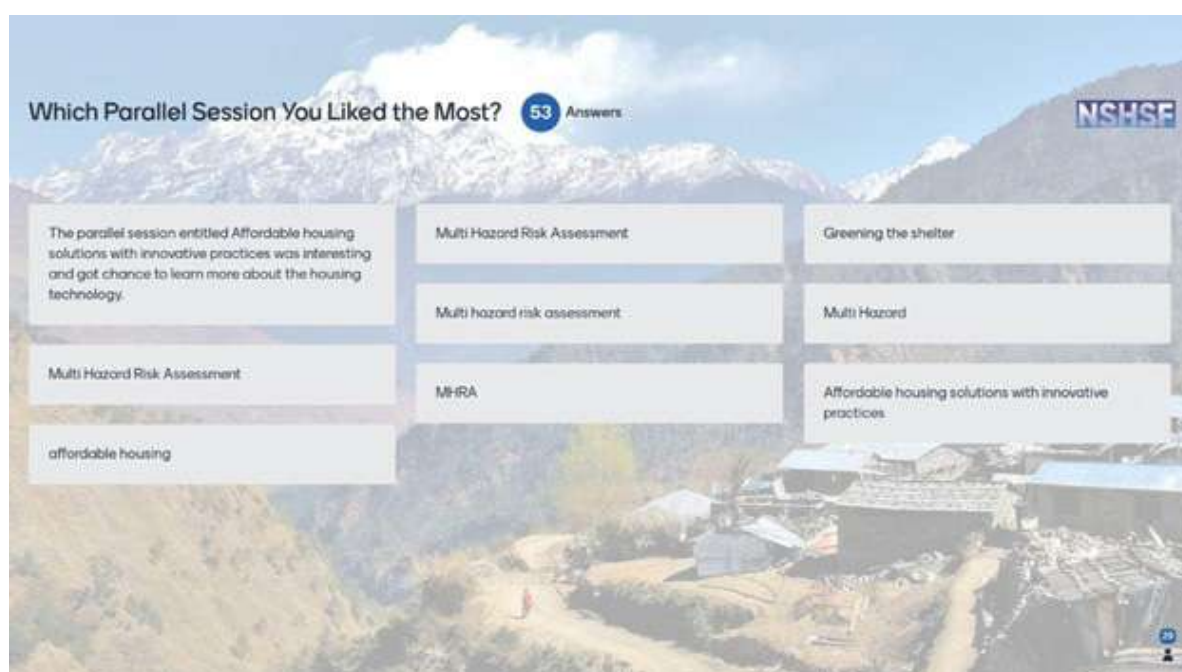
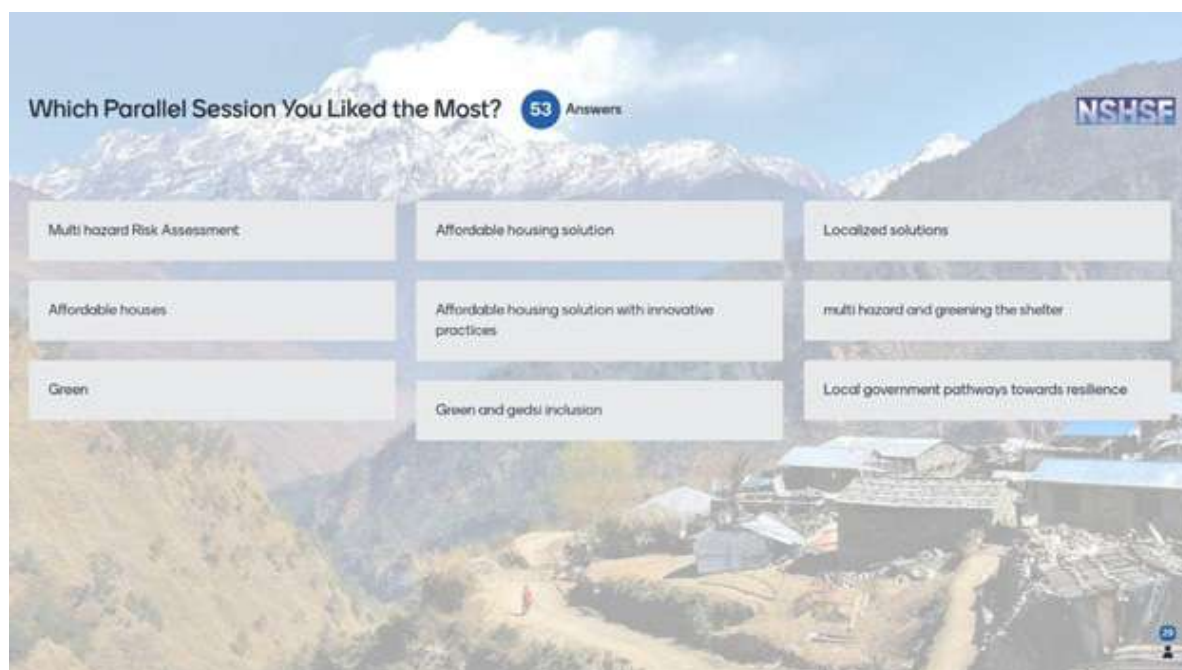
planning is recognized as a crucial aspect of disaster risk reduction.

Q. What are the challenges in the coming future and the path towards achieving the goal of shelter reconstruction, recovery, and disaster resilience in a united manner?

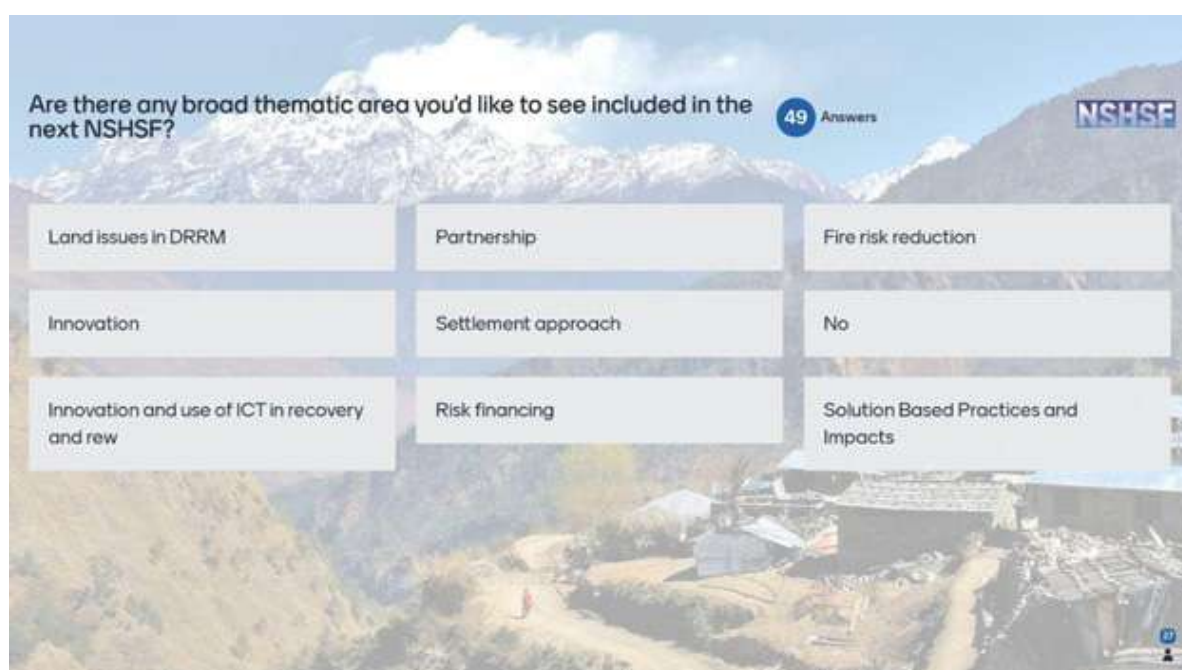
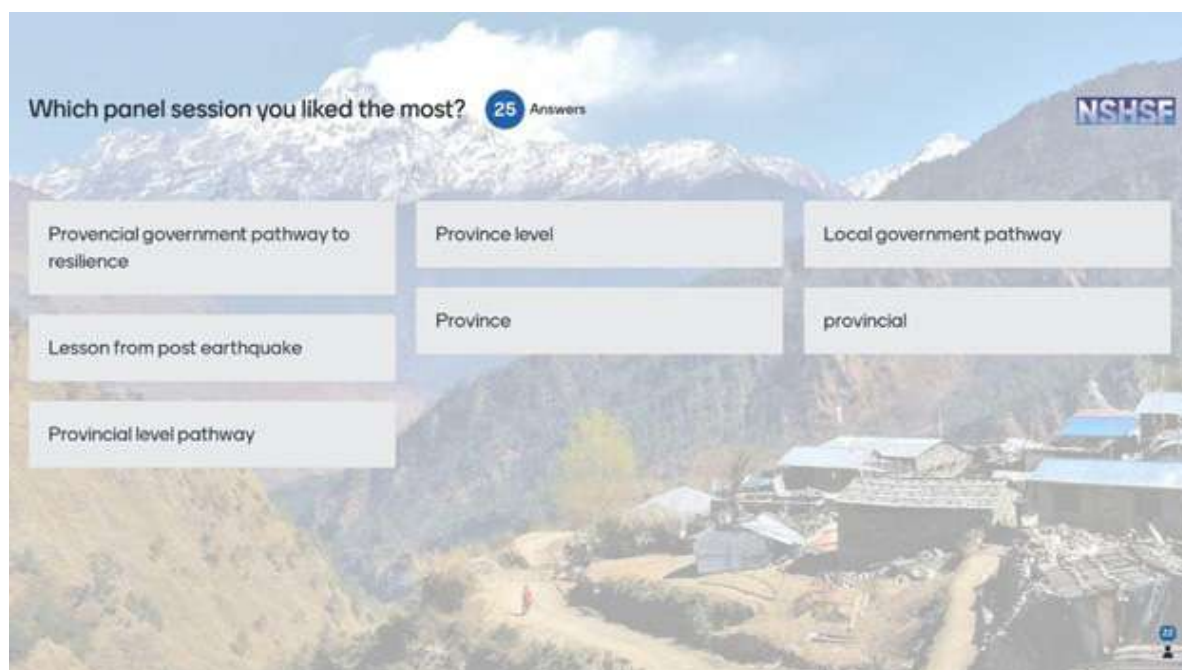
A. The challenges in the coming future include verification and adherence to building codes, distribution of relief items, shelter and geographical challenges, awareness and human resource capacity, limited resources and finance, coordination and collaboration, investment in disaster risk reduction, and thoughtful land management and planning. The path towards achieving the goal in a united manner involves fostering coordination, collaboration, and intersectoral coordination, investing in disaster risk reduction, empowering provincial governments, and clearly defining roles and responsibilities among stakeholders.

Feedback as received in Mentimeter



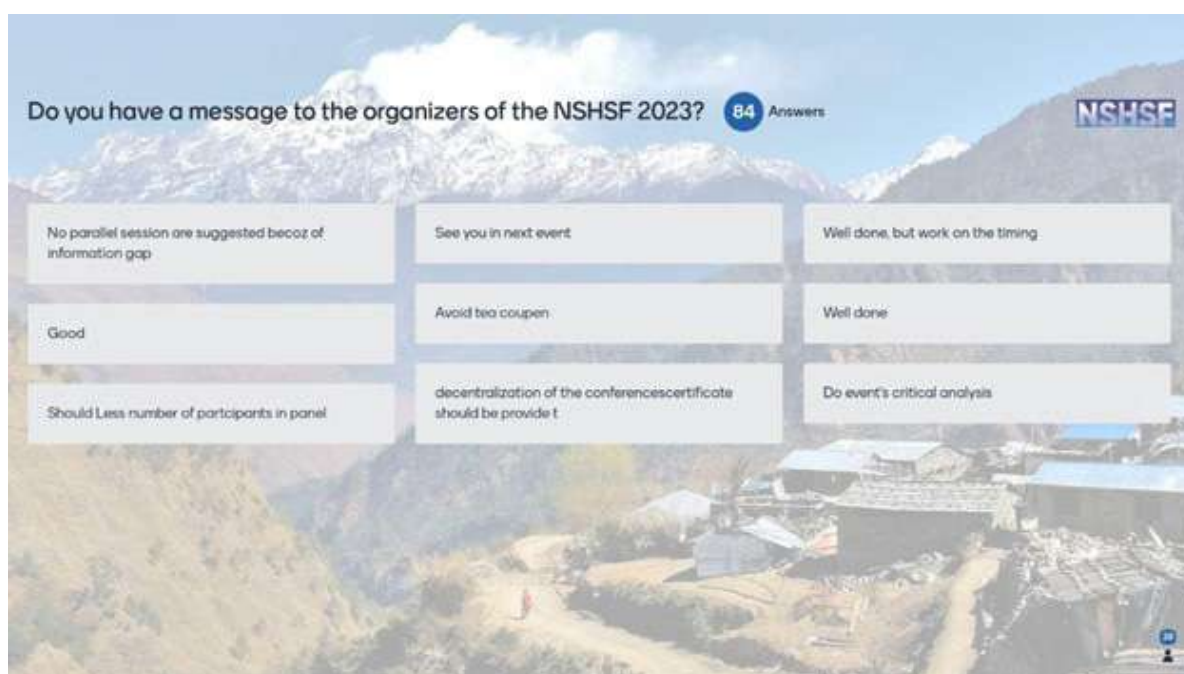


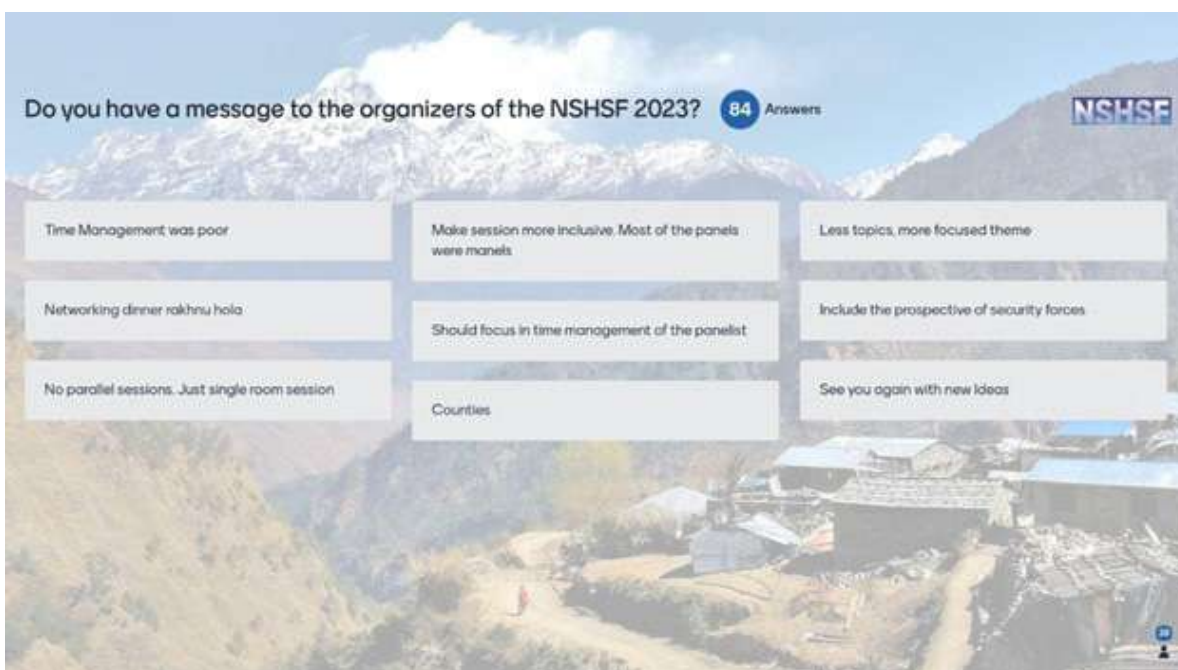
















What is your one major takeaway from the NSHSF 2023?

48 Answers

NSHSF

No more fancy screen in conference

Understanding from different level authorities

Ownership and collaboration between Govt (National, Province, Local), Civil society and Private sector

work of ismod, Sindh peoples approach, flood resilient housing solutions from pakistan, provincial ministry's efforts and constraints

Sharing and learning but few of the session should be standardized should provide the platform for new things, repeated conversation and dialogue should be identified and avoid in future.

Empowerment

A joint platform to share knowledge and link concerned stakeholders together

Data sharing and data partnership mechanism governments and stakeholders

Government plans and it's lacks. Do strengthening for policy making and it's implementation

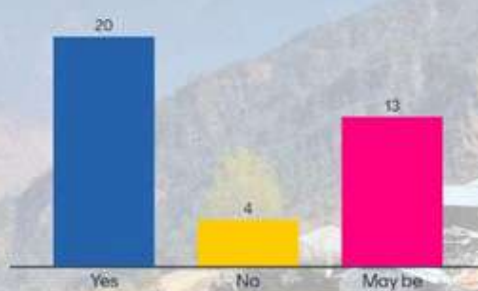
If you have participated virtually, how would you rate your experience?

NSHSF



Did NSHSF 2023 meet it's spirit to naturally improve coordination, collaboration, share learning & best practices?

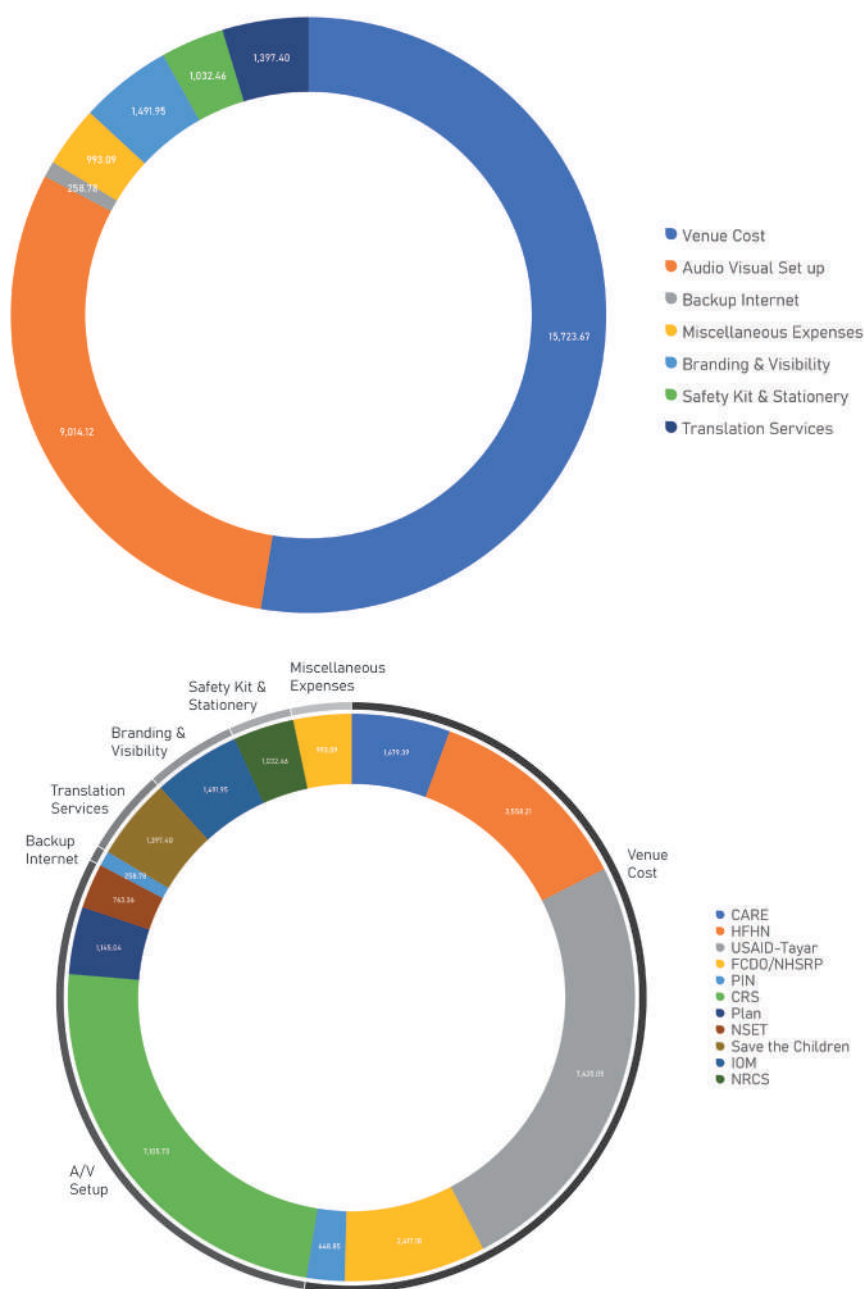
NSHSF



Budget Expenditure and Income Details

The total expense of 32,682.72 USD was made to organize the Asia Shelter Forum 2021. Total fund amounting of NPR 3,918,402.00 (approx. USD 30,000) were collected and contributed from the NSHSF 2023 Organizing Committee

Members (USAID, UKAid, CRS, NSET, HfH, PIN, Save the Children, CARE, Plan International, IOM, IFRC-NRCS). This doesn't include the management efforts laid by NSHSF Secretariat (NHSRP) and the organizing committee members' time allocated to design, plan and implement the event.



Terms of Reference - NSHSF 2023



National Disaster Risk Reduction and Management Authority (NDRRMA)

Nepal Shelter, Housing and Settlements Forum 2023

25 and 26 April 2023 (Baisakh 12 and 13)

Introduction:

The Nepal Shelter, Housing and Settlement Forum is the first of its kind in Nepal, although there have been numerous events held at various regional and individual country levels. The need of Nepal Forum was perceived in Asia Shelter Forum (ASF) in 2021.

The Asia Shelter Forum (ASF) is an assembly of Shelter and Settlement (S&S) practitioners founded by and for professionals involved in S&S programs across Asia. Its inaugural meeting in Bangkok in 2015 was followed by subsequent gatherings in Bangkok (2018), Indonesia (2019), and Nepal (2020-21). Each successive event has garnered a larger and more diverse audience, fostering an environment for the exchange of experiences and knowledge across Asia. The ASF strives to create a platform that cultivates professional connections and an understanding of best practices in shelter and settlement programming in the Asia Region. To promote broader participation by national and international practitioners, the ASF has actively pursued co-hosting opportunities with other regional or national S&S events. For instance, in 2015, ASF was given space by Habitat for Humanity as a side event to the Asia Pacific Shelter Forum, in 2018, CRS hosted the forum as a side event to their regional S&S meeting, and in 2019, it was organized by the Indonesian National Shelter Cluster as a side event to their National Shelter Workshop.

In 2020-21, the Nepal Government hosted the ASF as a hybrid event that attracted a record 574 participants from 87 organizations and independents from across the Asia-Pacific region and beyond. Additionally, top-level representatives from five governments attended the forum.

Nepal Shelter, Housing & Settlement Forum 2021 Declaration and Commitment:

The Nepal Day during Asia Shelter Forum 2021 was regarded as an opportunity for governments, I/NGOs, and community-based organizations (CBOs) to share and identify the most appropriate housing solutions for safe and dignified homes for vulnerable households through improved collaboration and networking. This Regional Forum boosted robust dialogues between governments, non-governmental organizations, Donor agencies, CBOs and practitioners to ensure sustainability and institutionalization of best practices. The event escalated the promotion of forming the National Shelter, Housing & Settlement Forum for collective coordination among shelter, Housing and DRR practitioners and government agencies to ensure local governments and communities are equipped with knowledge, systems and resources and empowered to lead timely and appropriate responses.

ToR Link: <https://bit.ly/3M6LDYw>

Program Schedule

Day 1: 25th April 2023



Day 1, 25 April 2023			Nepal Shelter Housing and Settlements Forum (NSHSF) 2023, Kathmandu, Nepal From Recovery to Resilience: Implementing Nepal's Declaration on Green, Resilient and Inclusive Development Pathway		
Lead Agency			National Disaster Risk Reduction and Management Authority (NDRRMA)		
Supporting Organization			USAID, UKAID, CRS Nepal, IFRC-NRCS, CARE Nepal, International Organization for Migration (IOM), Habitat for Humanity, Plan International, People in Need, Save The Children, National Society for Earthquake Technology Nepal (NSET), National Housing and Settlement Resilience Platform (NHSRP)		
Member Secretariate			National Housing and Settlement Resilience Platform (NHSRP)		
Start	End	Min	Session Name	Session Details	Facilitator
8:00	9:00	60	Breakfast/Registration	Breakfast and Registration	Mr. Rahul Pratap Singh, National Housing and Settlement Resilience Platform (NHSRP)
9:00	9:40	40	Inauguration	- Welcome Remarks by Mr. Rameshwar Marahatta, Joint Secretary, National Disaster Risk Reduction and Management Authority (NDRRMA) - Remarks by Mr. Surendra Mohan Shrestha, Director General, Department of Urban Development and Building Construction (DUDBC) - Remarks by Hon. Minister Sita Gurung, Ministry of Urban Development (MoUD) - Remarks by Session Chair: Mr. Rudra Singh Tamang, Joint Secretary, National Disaster Risk Reduction and Management Authority (NDRRMA)	Ms. Ranjana Shrestha, Communication Expert, National Disaster Risk Reduction and Management Authority (NDRRMA)
9:40	10:00	20	Household & Local Government Testimonials	- Ms. Dev Kumari Pariyar, 2015 Earthquake Affected Household, Ramechhap Municipality - Mr. Arjun Gurung, Chairperson, Marsyangdi Rural Municipality, Lamjung	Ms. Preeti Karki, Technical Coordinator, National Housing and Settlement Resilience Platform (NHSRP)
10:00	11:15	75	Panel: Lesson Learned & Way Forward for a Green, Multi-hazard Resilient and Inclusive Recovery	- Mr. Damodar Gautam, Director General, Department of Archeology - Mr. Bharat Aryal, Director General, Department of Local Infrastructure (DoLI) - Mr. Kosh Nath Adhikari, Project Director, CLPIU - Building and Housing - Mr. Dilli Luitel, Project Director, CLPIU - Education - Mr. Krishna Gaire, Under Secretary, National Disaster Risk Reduction and Management Authority (NDRRMA)	Mr. Surya Narayan Shrestha, Executive Director, National Society for Earthquake Society Nepal (NSET)
11:15	11:30	15	Pakistan Nepal Collaboration for Resilient Recovery and Climate Adaptive Solutions	- Ms. Sana Khawaja, Communication Lead & Technical Advisor, Sindh People Housing for Flood Affected (SPHF) - Mr. Minar Thapa Magar, National coordinator, National Housing and Settlement Resilience Platform (NHSRP)	Ms. Preeti Karki, Technical Coordinator, National Housing and Settlement Resilience Platform (NHSRP)
11:30	12:30	60	Panel: Lessons from 2015 Earthquake Recovery	- Mr. Surya Narayan Shrestha, Executive Director, National Society for Earthquake Society Nepal (NSET) - Mr. Taranidhi Bhattarai and Mr. Kishor Thapa on 'Book - Bahattar Saalko Bhukampa' - Mr. Chandra Bahadur Shrestha, Former Executive Committee Member, NRA - Mr. Minar Thapa Magar, National Coordinator, NHSRP	Mr. Sushil Gyawali, Former CEO of National Reconstruction Authority (NRA)
12:30	13:30	60	Lunch and Market stall	Discussion with participants and Market Stall over Lunch	Mr. Rahul Pratap Singh, National Housing and Settlement Resilience Platform (NHSRP)
			Parallel Presentations A: Main Hall DRR and Recovery from Multiple Hazards	- Minimum Reconstruction Guidelines for Flood Resilient Houses Safe-Environmentally Friendly-Affordable, Mr. Muhammad Naveed, Engineering Team lead, Government of Pakistan - Shock Responsive Social Protection in DRRM Governance, Mr. Hemanta Dangal, Save the Children Nepal - Post-Earthquake Reconstruction and Recovery of Private Residential Buildings following Doti Earthquake, Mr. Rabin Chaulagain, NDRRMA - NHSRP - Enhancing Disaster Management Process in Nepal through Good Governance: Best Practices for Effective Response and Recovery, Mr. Khagendra Awasthi, GRDSC	Chair: Mr. Krishna Gaire, Under Secretary, NDRRMA Facilitator: Mr. Rabin Dahal, Save the Children Nepal
			Parallel Presentations B: Kirtipur Hall Localization Lessons from 2015 Earthquake Recovery	- Learnings from Post-Earthquake Housing Reconstruction, Mr. Ranjan Dhungel, Director, National Society for Earthquake Society Nepal (NSET) 'Baliyo' Build Back Better: An inquiry into the Choice of Construction Materials for Housing Reconstruction in Gyalchowk Households, Mr. Tobias Voluhl - Stakeholders Coordination for Localizing Seismic Resilience, Mr. Krishna Prasad Dhital, Chairperson, Rural Awareness and Development Organization Nepal - Nepal Disaster Resilience Localization process for Reconstruction and Settlement Recovery, Mr. Rijan Gajurel, National Housing and Settlement Resilience Platform (NHSRP)	Chair: Mr. Kosh Nath Adhikari, Project Director, CLPIU - Building Facilitator: Ms. Reshma Shrestha, CRS
14:50	15:20	30	Tea Break	Discussion with Participants over Tea (Market Stall Visit)	Mr. Rahul Pratap Singh, National Housing and Settlement Resilience Platform (NHSRP)
			Parallel Presentations A: Main Hall DRR and Recovery from Multiple Hazards	- Use of Multi-hazard Scenario Ensembles to Inform Better Decision Making and Reduce the Impacts of Multi-hazards in Nepal, Ms. Sweeta Sijapati, SAJAG Program - NSET - Shelter and Housing Pre-Crisis Market Analysis, Mr. Rupendra Basnet, Habitat for Humanity - Multi-Hazard Risk Informed City Planning, Ms. Bandana Sharma, USAID-Tayar - Adaptation Solutions: Flood Resilient Habitat and Heat Stress Management in Urban Housing, Ms. Erica Udas, Ecosystem Specialist, ICIMOD	Chair: Mr. Sundar Sharma, Under Secretary, NDRRMA Facilitator: Mr. Rabin Chaulagain, National Housing and Settlement Resilience Platform (NHSRP)
			Parallel Presentations B: Kirtipur Hall Localization Lessons from 2015 Earthquake Recovery	- Municipal Led Safe and Affordable Housing Program, Mr. Raj Kumar Das, Chief Administrative Officer, Shambhunath Municipality - DRRM Localization Training: Lessons Learned and Way Forward, Ms. Ruchi Thapa, Project Lead, IOM - People/Community Centered Approaches, Mr. Manish Timsina, Technical Coordinator, NRCS/IFRC - Municipal Collaboration with Partners- Experience working with CityNet, Bishnu Paudel, Planning Officer, Tansen Municipality	Chair: Mr. Rishi Raj Acharya, Under Secretary, Ministry of Federal Affairs and General Administration (MoFAGA) Facilitator: Mr. Manish Timsina, IFRC-NRCS
16:40	16:45	5	Day 1 Feedback	Participants Share Appreciation and Recommendations through Menti meter (Live Poll)	Mr. Rijan Gajurel, Technical Coordinator, National Housing and Settlement Resilience Platform (NHSRP)
16:45	16:55	10	Closing Session	Mr. Rudra Singh Tamang, Joint Secretary, NDRRMA (closing remarks)	

NSHSF 2023

Day 2: 26th April 2023

Day II, 26 April 2023			Nepal Shelter Housing and Settlements Forum (NSHSF) 2023, Kathmandu, Nepal From Recovery to Resilience: Implementing Nepal's Declaration on Green, Resilient and Inclusive Development Pathway		
Lead Agency			National Disaster Risk Reduction and Management Authority (NDRRMA)		
Supporting Organization			USAID, UKAID, CRS Nepal, IFRC-NRCS, CARE Nepal, International Organization for Migration (IOM), Habitat for Humanity, Plan International, People in Need, Save The Children, National Society for Earthquake Technology Nepal (NSET), National Housing and Settlement Resilience Platform (NHSRP)		
Member Secretariate			National Housing and Settlement Resilience Platform (NHSRP)		
Start	End	Min	Session Name	Session Topic	Facilitator
8:00	8:45	45	Breakfast/Registration	Breakfast and Registration	Mr. Rahul Pratap Singh, National Housing and Settlement Resilience Platform (NHSRP)
8:45	9:00	15	Day Overview	Second day overview and highlights	Ms. Preeti Karki, Technical Coordinator, NHSRP
9:00	10:20	80	Parallel Presentations A: Main Hall Greening Shelter and Sustainable Housing Practices	- Greening Shelter Reconstruction, Ms. Preeti Karki and Mr. Laxmi Prasad Bhatta, National Housing and Settlement Resilience Platform (NHSRP) - Bamboo Housing for Climate Resilience and Sustainable Living, Mr. Pranab Pradhan, Green Bamboo Creation Private Limited - Low-cost Climate Friendly Houses through Micro Enterprises, Mr. Bjorn Soderberg, Build Up Nepal - Approaches for Energy and Resource Efficient Buildings, Mr. Dibya Acharya, Sr. Architect, MinErgy - Environmental Country Profile, Mr. Ajay Kumar KC, Water Sanitation and Environment Specialist- Consultant, NRCS/IFRC	Chair: Ms. Mira Gyawali, Under Secretary, DUDBC Facilitator: Ms. Preeti Karki, NHSRP
			Parallel Presentations B: Kirtipur Hall Integrating Gender, Disability and Social Inclusion (GEDSI) in DRRM	- Integrating Gender, Disability and Social Inclusion (GEDSI) in DRRM: Sharing of Draft National Strategic Action Plan, Ms. Reena Chaudhary, GEDSI Focal Person, NDRRMA - Building Resilience through Inclusive and Climate-Action Disaster Risk Reduction, Ms. Januka Neupane, Bagmati Province Coordinator, WHDRRP - Durable Local Solutions Approach for Inclusion and Resilience, Ms. Arishma Shrestha, GESI and Protection Lead, People In Need - Integrated Approach for Shelter Housing and Settlements, Mr. Bhushan Aryal, Construction Specialist, Plan International	Chair: Mr. Rudra Singh Tamang, Joint Secretary, NDRRMA Facilitator: Mr. Rupendra Basnet, Habitat for Humanity
			Parallel Presentations C: Kirtipur Hall Multi-Hazard Risk Assessment	- Multi-hazard Risk Assessment - Experience from NDRRMA - Mr. Surya Narayan Shrestha, Executive Director, NSET - Mr. Uttam Paudel, Technical Advisor DRR, CRS - Mr. Kusang Sherp, Department of Irrigation Landslide Risk Assessment, Department of Mines and Geology	Session Chair: Mr. Anil Pokhrel, Chief Executive, NDRRMA Facilitator: Mr. Pradip Khatiwad, YIL
10:20	11:40	80	Parallel Presentations A: Main Hall Affordable Housing Solutions With Innovative Practices	- Green and affordable building materials for housing construction, Mr. Shuva Sharma, Professional Lead, Scott Wilson Nepal - Affordable Housing Solutions, Mr. Suraj Shrestha, DRRM Coordinator, CARE Nepal - Affordable Housing Finance Solutions to Address Housing needs of Low-Income Communities, Ms. Lajana Manandhar, Executive Director, Lumanti - Lime Stabilized Soil Initiative-Field Experience from Banke District, Mr. Rupesh Shrestha, Technical Coordinator, CRS Nepal - Shelter Solution through Housing Microfinance, Mr. Sunil Khanal, Sr. Manager, Habitat for Humanity	Chair: Mr. Ram Chandra Dangal, Joint Secretary, Ministry of Urban Development (MoUD) Facilitator: Mr. Laxmi Prasad Bhatta, NHSRP
			Parallel Presentations B: Kirtipur Hall Integrating WASH, Livelihoods, Land For Building Resilient Homes and Communities	- Integration of Water Scheme, Livelihood and Shelter Component for Building Resilient Homes and Communities, Mr. Mehul Savla, Program Manager, CRS Nepal - Indigenous Knowledge and Practice, Mr. Gajendra Thapaliya, Chief Administrative Officer, Jhapa Rural Municipality - Innovative and Green Housing Practices, Ms. Salina Pradhan, Architect, Habitat For Humanity - The Impacts of Earthquake Disaster on Educational Accessibility, Affordability and Continuity in Nepal, Mr. Nirmal Chongbang, PhD Student, Tribhuvan University	Chair: Mr. Chandra Shiwakoti, Under Secretary, MoLCPA Facilitator: Ms. Kamala KC, British Embassy Kathmandu- Foreign, Commonwealth & Development Office (FCDO)
11:40	12:00	20	Video Documentary: Lessons from 2015 Earthquake Recovery	Government Video Documentary for the Lessons Learned from the 2015 Earthquake Recovery	Ms. Preeti Karki, Technical Coordinator, NHSRP
12:00	13:00	60	Lunch	Discussion with Participants over Tea (Market Stall Visit)	Mr. Rahul Pratap Singh, NHSRP
13:00	14:15	75	Panel: Provincial Government Pathway towards Resilience	- Mr. Dhurba Gaida, Secretary, MoAL, Bagmati Province - Mr. Dharma Raj Ojha, Under Secretary, Land Management, Agriculture and Cooperative, Sudurpaschim Province - Mr. Yam Bahadur Kunwar, Under Secretary, MoAC, Lumbini Province - Mr. Suran Prasad Sharma Paudel, Under Secretary, MoAL, Gandaki Province - Mr. Umesh Basnet, Secretary, MoAL, Koshi Province - Mr. Nidhi Raj Neupane, Under Secretary, MoAL, Karnali Province	Chair: Mr. Pradip Koirala, Joint Secretary, MOHA Facilitator: Mr. Minar Thapa Magar, NHSRP
14:15	14:45	30	Tea Break	Discussion with Participants over Tea (Market Stall Visit)	Mr. Rahul Pratap Singh, NHSRP
14:45	16:00	75	Panel: Local Government Pathway towards Resilience	- Ms. Laxmi Devi Pande, Chairperson, NARMIN - Mr. Jit Narayan Thapa Magar, Chairperson, Budhiganga Rural Municipality, Morang - Mr. Jaya Narayan Shah, Chairperson, Jhapa Rural Municipality, Jhapa - Mr. Kala Nidhi Devkota, Executive Director, MuAN - Mr. Bishnu Paudel, General Secretary, CNNC	Mr. Rudra Singh Tamang, Joint Secretary, NDRRMA Facilitator: Mr. Reshma Shrestha, CRS
16:00	16:05	5	Day II Feedback	Participants Share Appreciation and Recommendations through Mentimeter (Live Poll)	Mr. Rijan Gajurel, Technical Coordinator, NHSRP
16:05	17:05	60	Closing Session	- Remarks from Special Guest: Mr. Binod Prakash Singh, Secretary, MoHA - Closing of the Two Days Event, Mr. Anil Pokhrel, Chief Executive, NDRRMA - Remarks from Special Guest: Dr. Ganga Lal Tuladhar, Council Member, NDRRMA - Appreciation and Token of Love Distribution by Chief Guest, Hon'ble Mr. Narayan Kaji Shrestha, Deputy Prime Minister and Home Minister - Closing Remarks by Chief Guest, Hon'ble Mr. Narayan Kaji Shrestha, Deputy Prime Minister and Home Minister	Ms. Ranjana Shrestha, Communication Expert, National Disaster Risk Reduction and Management Authority (NDRRMA)

NSHSF 2023

1st National Conference



Nepal Shelter Housing and Settlements Forum
Kathmandu, 25-26 April 2023

From Recovery to Resilience: Implementing Nepal's Declaration on Green, Resilient, and Inclusive Development Pathway

Organized by:



Government of Nepal
Ministry of Home Affairs
National Disaster Risk Reduction and Management Authority
Singhadurbar, Kathmandu

Supported by:



Save the Children

